

# Chapter 6: Geospatial Data Visualization

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# Sources of Geospatial Data

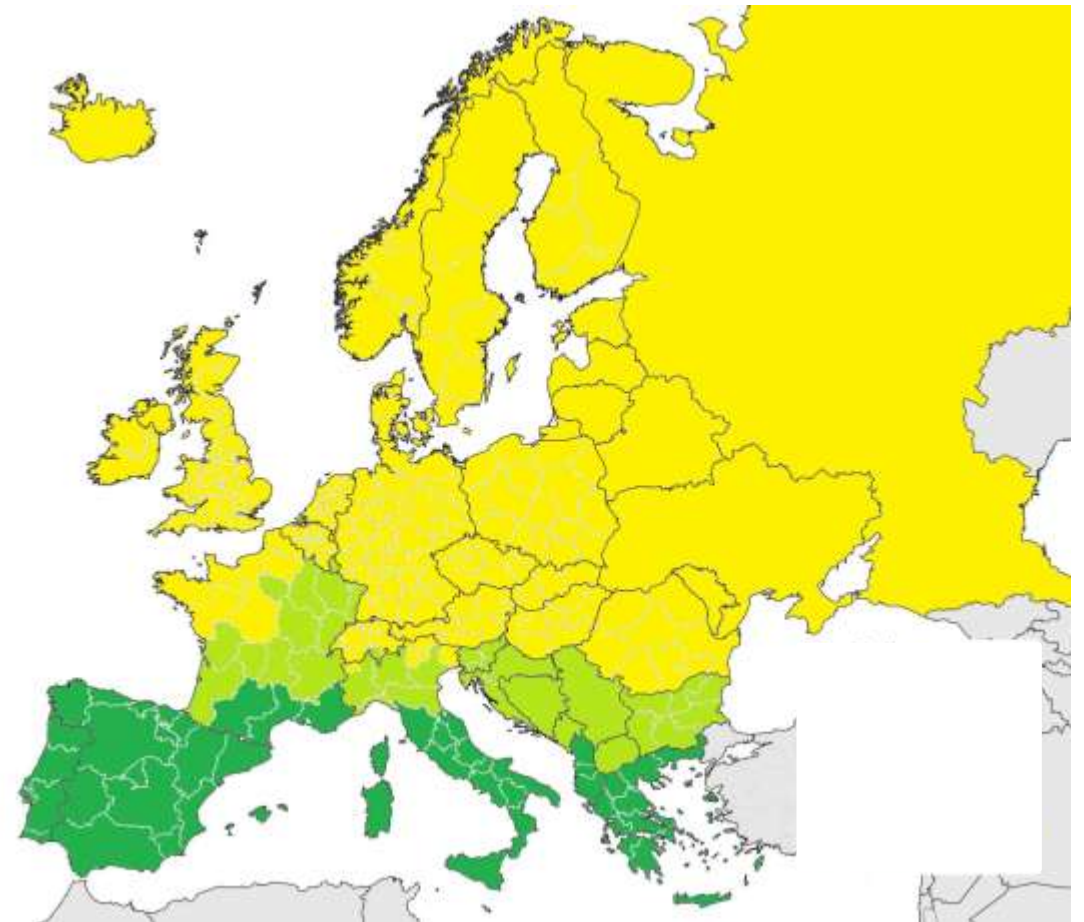
- **Geospatial data** is linked to specific locations on earth
- It arises in many application contexts. *Examples* include:
  - Economic and social indicators (unemployment rates, crime rates)
  - Weather phenomena (temperatures, winds, rainfall)
  - Traffic
  - Credit card payments
- Geospatial data is typically visualized with **maps**
  - Mapped items often have nonspatial attributes in addition to their spatial coordinates

# Thematic Maps

- **Thematic maps** visualize the geographic pattern of a specific subject (theme)

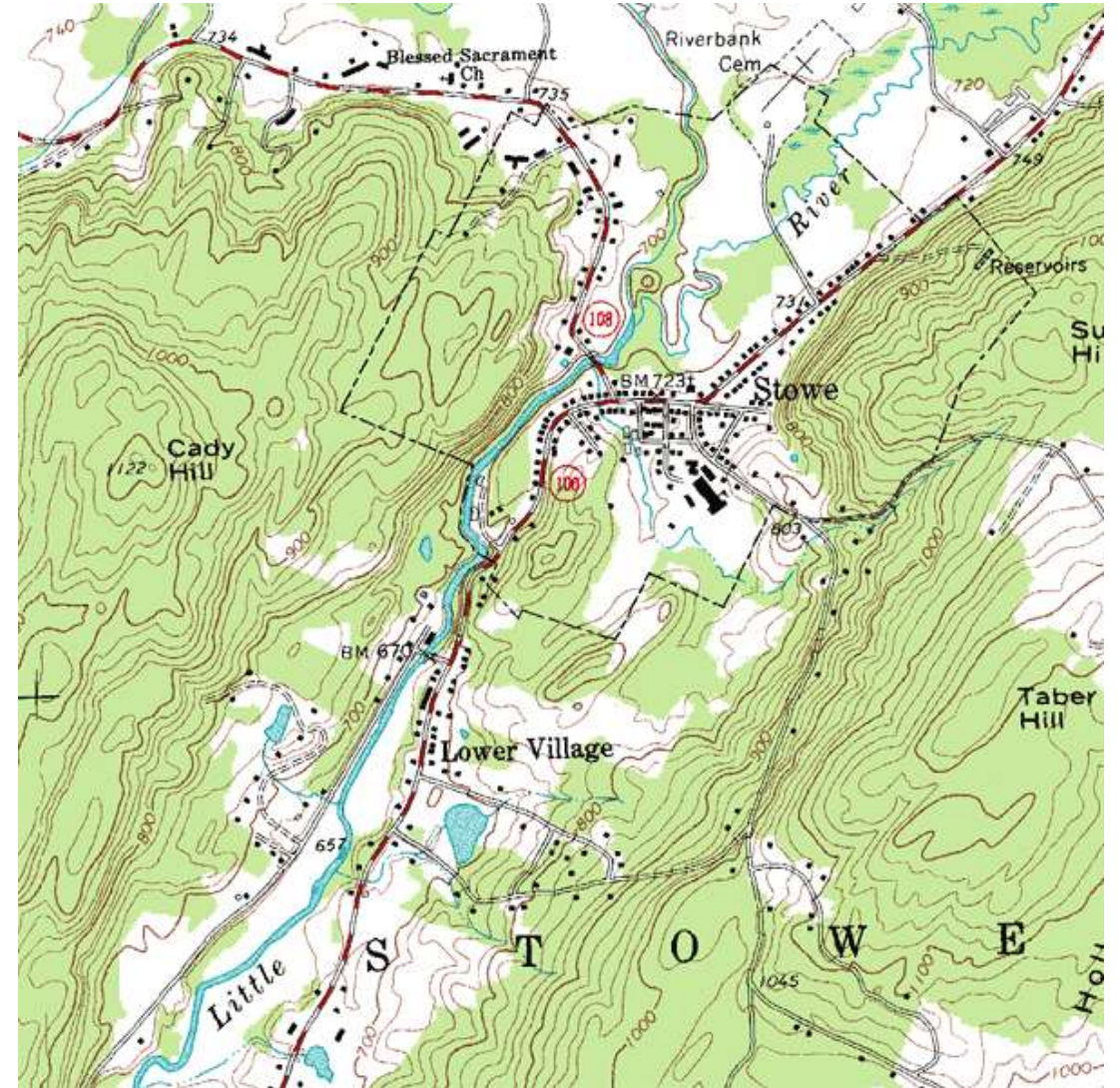
– *Examples:*

- Election results
- Migration patterns
- Production of types of goods
- Amount of trade between countries
- ...



# Topographic Maps

- **Topographic maps** display features of a local landscape
  - Relief
  - lakes and rivers
  - forest cover
  - buildings
  - roads and railways
  - ...



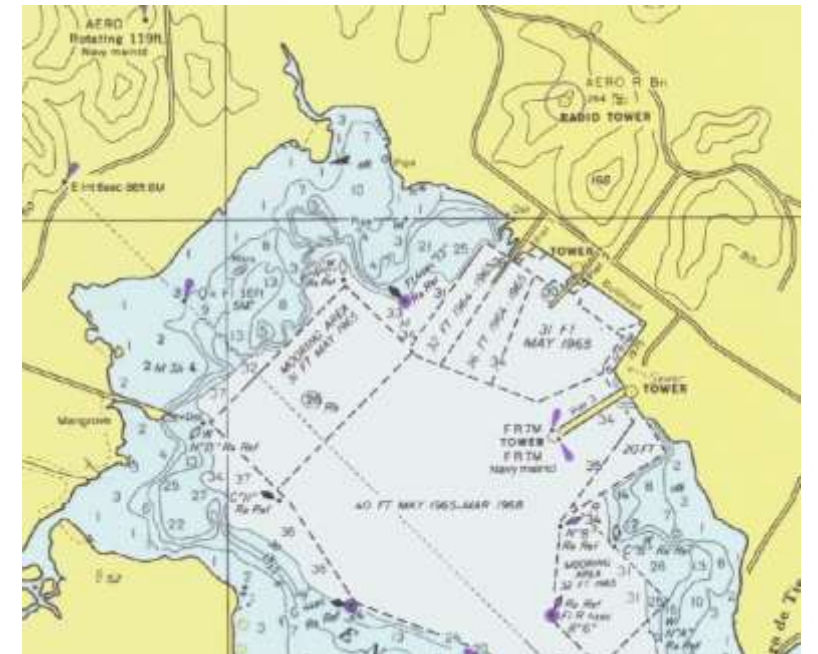
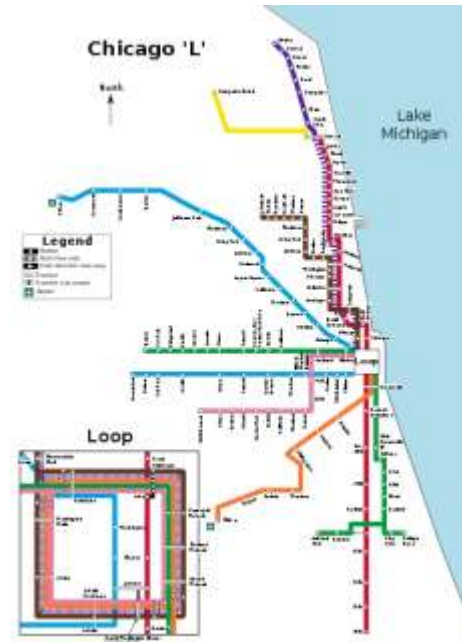


# Navigational Maps and Charts

- **Navigational maps and charts** focus on information that is relevant for navigation
  - **Maps** (e.g., road maps, subway maps) focus on land
  - **Charts** focus on water / marine navigation



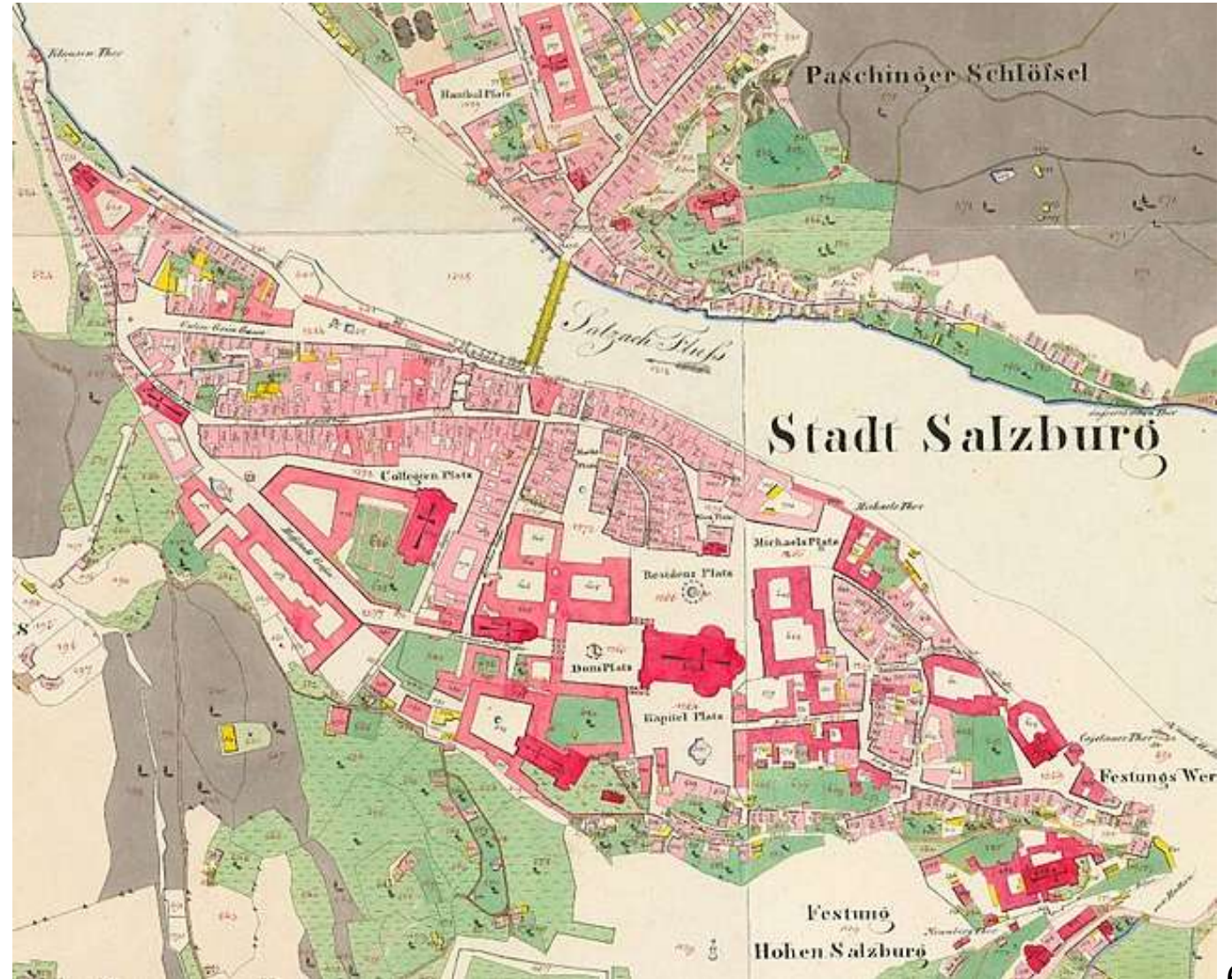
Touristic map of Paris (tripomatic.com)





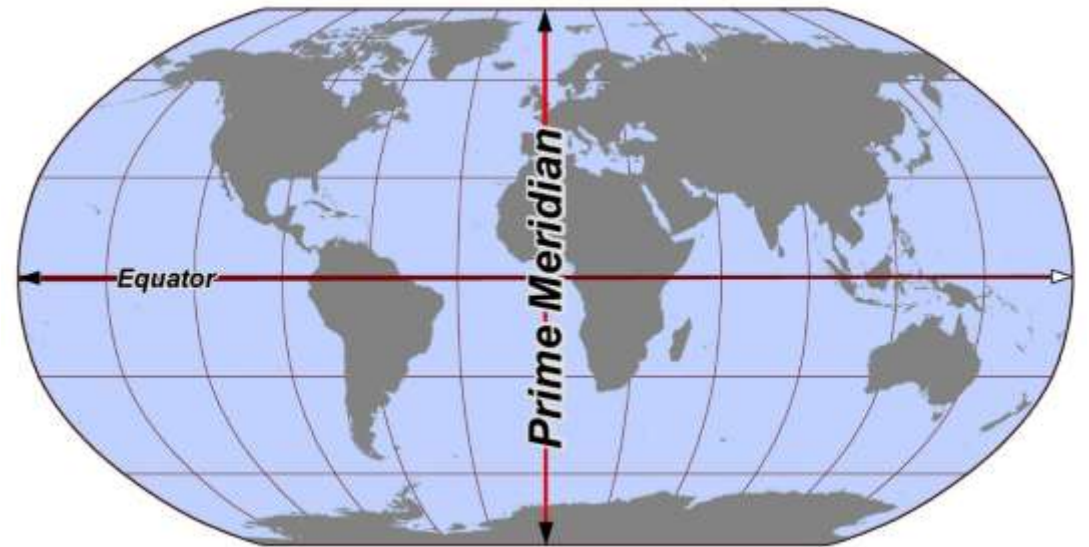
# Cadastral Maps

- **Cadastral maps** illustrate the parcels into which land is subdivided with respect to ownership and land use



# Conventions and Terminology

- For the purpose of map making, the (irregularly shaped) earth is often approximated as an ellipsoid or sphere
- Positions on the surface are specified as angles
  - **Latitude**  $\phi \in [-90^\circ, 90^\circ]$ 
    - Parallel = line of equal latitude
    - Equator ( $\phi = 0^\circ$ ) halfway between the poles
    - Negative  $\phi$  south, positive  $\phi$  north
  - **Longitude**  $\lambda \in [-180^\circ, 180^\circ]$ 
    - Meridian = line of equal longitude
    - Prime meridian ( $\lambda = 0^\circ$ ) close to the Royal Observatory in Greenwich
    - Negative  $\lambda$  west, positive  $\lambda$  east



# Specifying Positions More Exactly

- Traditionally, each degree is subdivided into 60 minutes, and each minute is subdivided into 60 seconds
  - *Example:* CS building at (50°43'38''N 7°5'8''E)
- Geographic Information Systems (GIS) more commonly use a decimal representation
  - *Example:*

$$50^{\circ}43'38'' = 50 + \frac{43}{60} + \frac{38}{3600} \approx 50.727^{\circ}$$

$$7^{\circ}5'8'' = 7 + \frac{5}{60} + \frac{8}{3600} \approx 7.086^{\circ}$$



# Types of Geospatial Data

- Geospatial data can be classified by its spatial **extent**:
  - **Point** phenomena have a location, but no relevant extent
    - Can be specified by a single longitude and latitude (plus attributes)
    - *Examples*: Location of transactions, buildings, cities
    - Point data is often aggregated at an area level
  - **Line** phenomena have length, but no relevant width
    - Can be specified by a series of longitude and latitude pairs (plus attributes)
    - *Examples*: Roads, boundaries between countries

# Types of Geospatial Data

- Geospatial data can be classified by its spatial **extent**:
  - **Area** phenomena have relevant two-dimensional extent
    - Can be specified by a series of longitude and latitude pairs that completely enclose a region (plus attributes)
    - *Examples*: Political units (e.g., countries), lakes, parks
  - **Surface** phenomena have relevant „two-and-a-half“-dimensional extent
    - Can be specified by longitudes, latitudes, and heights (plus attributes)
    - *Example*: Height fields / terrains
- Geospatial data often also has a **temporal** dimension
  - *Example*: Changes in weather or traffic conditions

# **Section 6.1: Map Projections**



# Properties of Map Projections

- **Maps** provide a planar representation of the earth
  - We will use a spherical approximation of the earth as a starting point
  - Map projection:  $\Pi: (\lambda, \phi) \rightarrow (x, y)$
- **Conformal projections** locally retain angles and shapes. They necessarily distort areas.
- In **equivalent / equal area projections**, regions cover the same area on the map as they do on the globe. They necessarily distort angles.
- **Gnomonic projections** map great circles to straight lines (straight lines are geodesics)



# Types of Map Projections

- Many map projections start by projecting the earth onto a **developable surface** which can then be flattened without further distortions
  - *Most commonly used:* (Pseudo-)cylindric projections



**Cylinder Projection**



**Plane Projection**



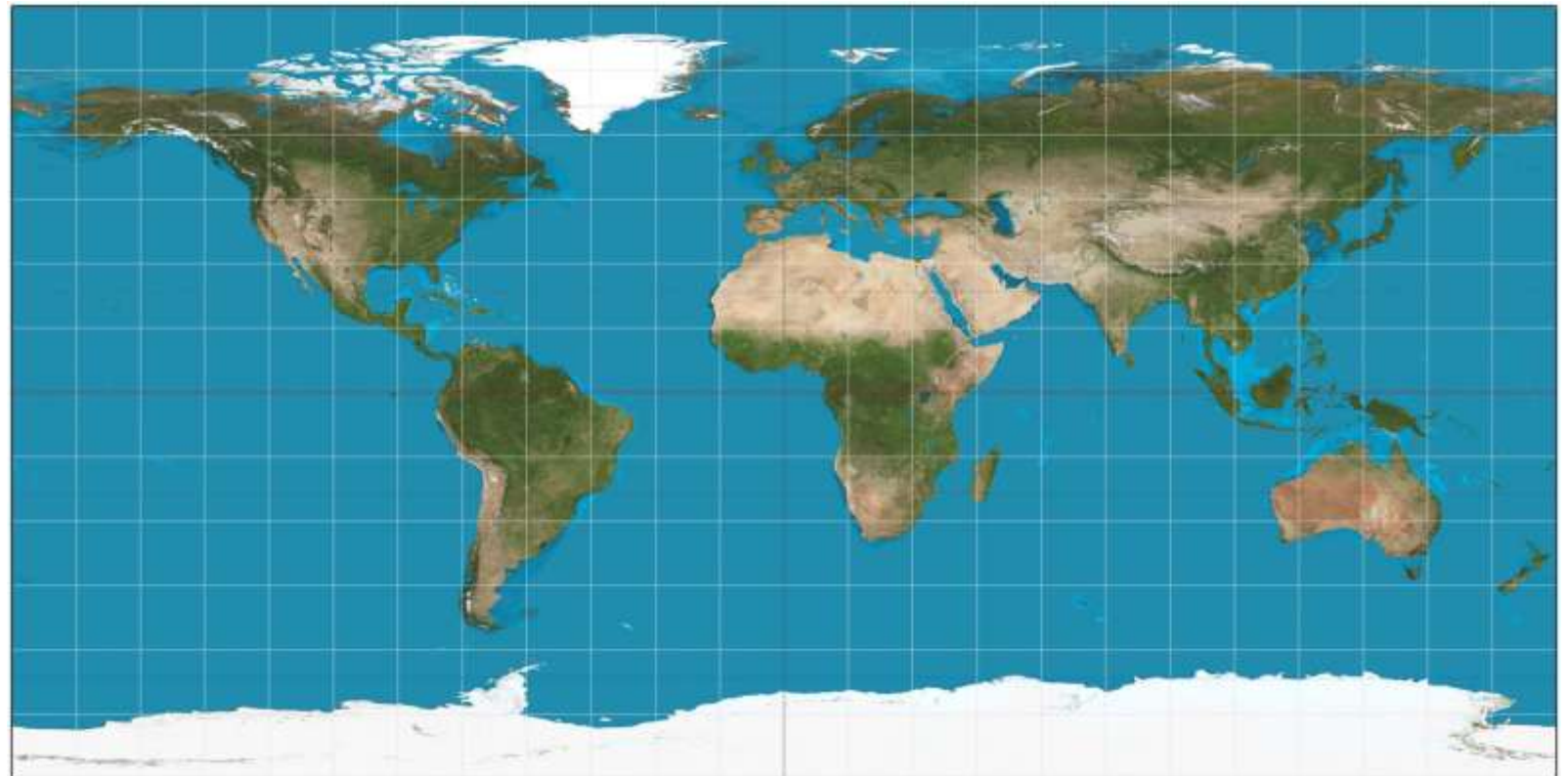
**Cone Projection**

# Equirectangular Cylindrical Projection

- The **equirectangular mapping** simply sets

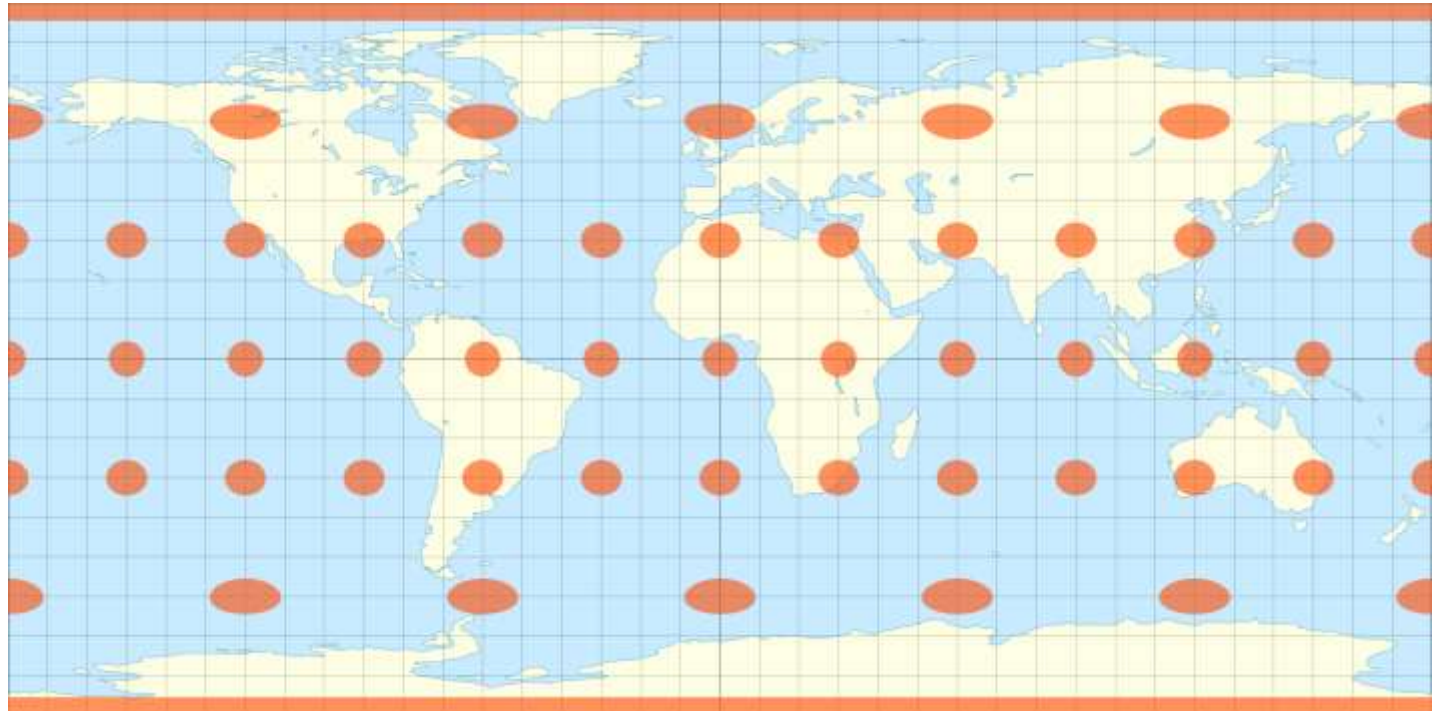
$$x := \lambda, y := \phi$$

- Easy to read off geographic location, can be used for thematic maps
- Neither conformal nor equal area



# Tissot's Indicatrix

- **Tissot's indicatrix** can be used to visualize the distortion due to a map projection
  - Ellipsoid that results from projecting a circle of infinitesimal radius
  - Maps distort angles when the indicatrix is not a circle
  - Maps distort areas when indicatrices' areas vary across the map



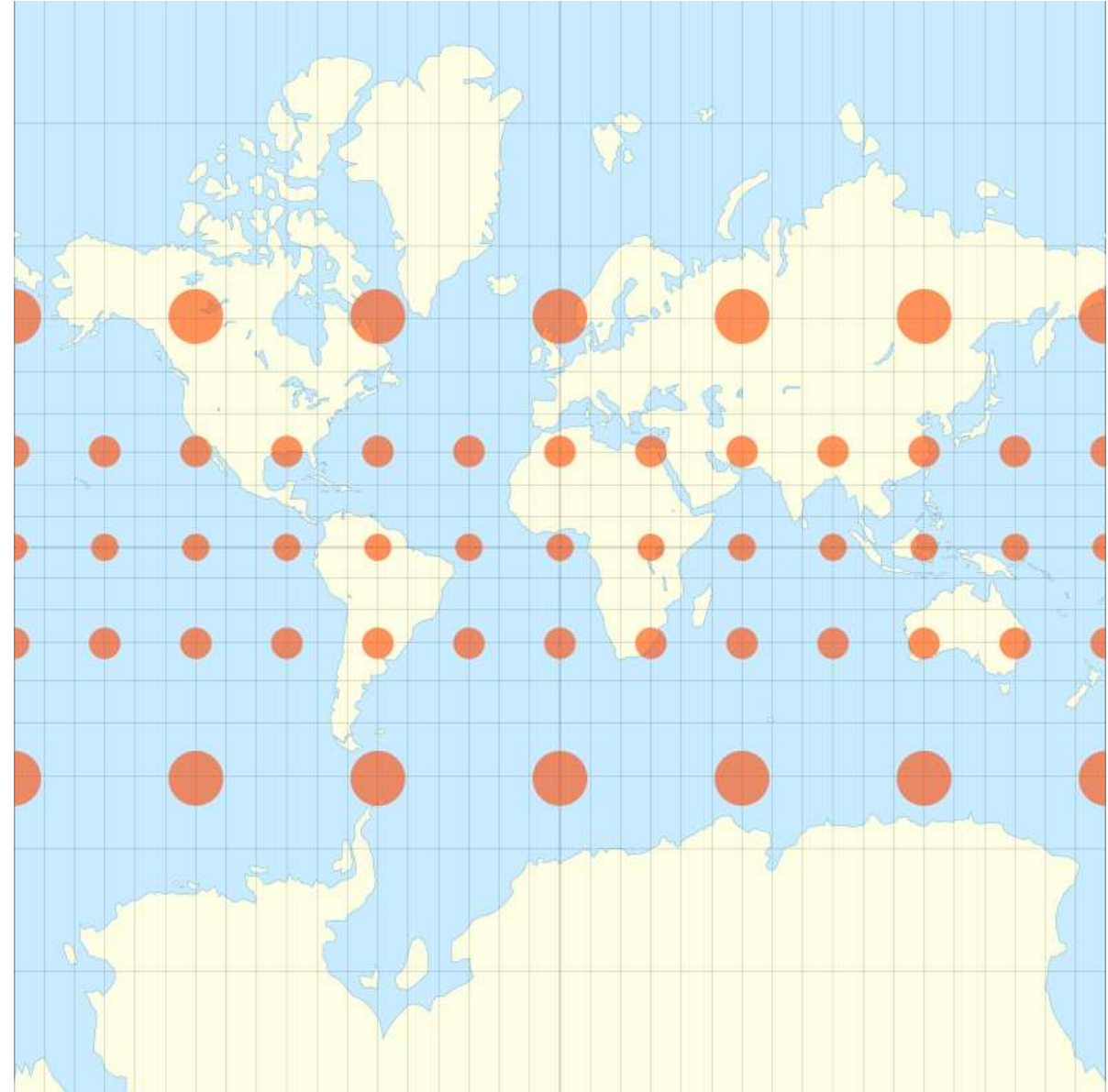


# Mercator Projection

- The **Mercator projection** is a conformal cylindrical projection

$$x := \lambda, y := \ln \left( \tan \left( \frac{\pi}{4} + \frac{\phi}{2} \right) \right)$$

- Requires truncating range of  $\phi$
- Popular for navigation
- Areas away from the equator are strongly inflated



# Illustrating a Mercator-based Misconception

## The True Size of Africa

Comparison of Africa's land area to selected countries/regions

African continent **30.4m km<sup>2</sup>**



\* excluding Russia.

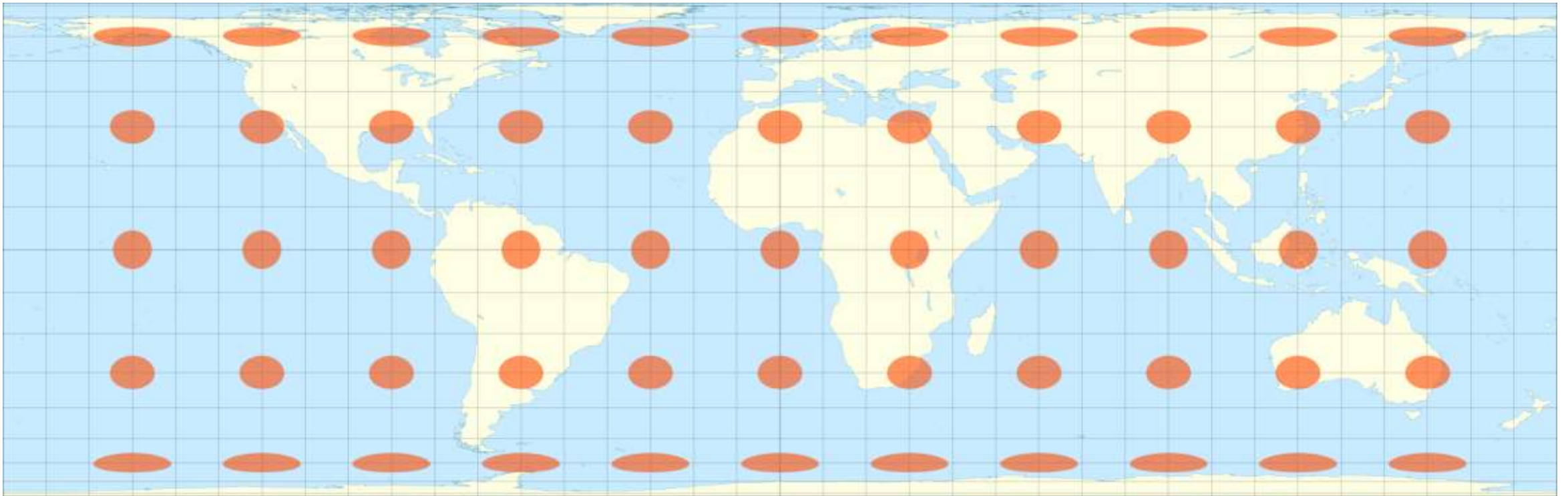
Sources: CIA, World Population Review, Visual Capitalist



# Lambert Cylindrical Projection

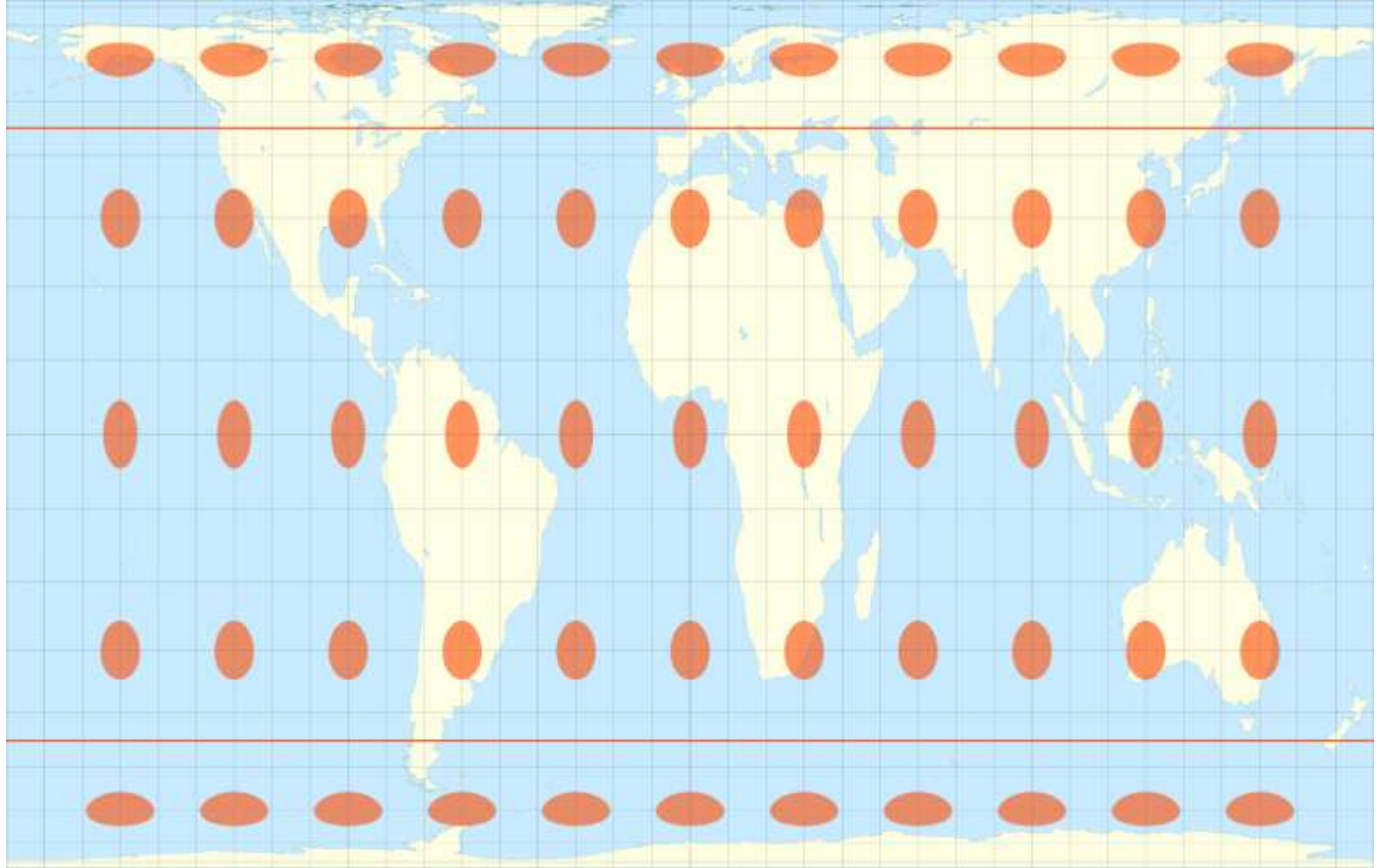
- The **Lambert cylindrical projection** is an equal-area projection that is undistorted along the equator

$$x := \lambda, y := \sin \phi$$



# Gall-Peters Projection

- The **Gall-Peters projection** is an equal-area projection that is undistorted along  $\phi = \pm 45^\circ$ 
  - Can be written as  $x := \lambda, y := 2 \sin \phi$
  - Reduces overall distortion compared to Lambert projection





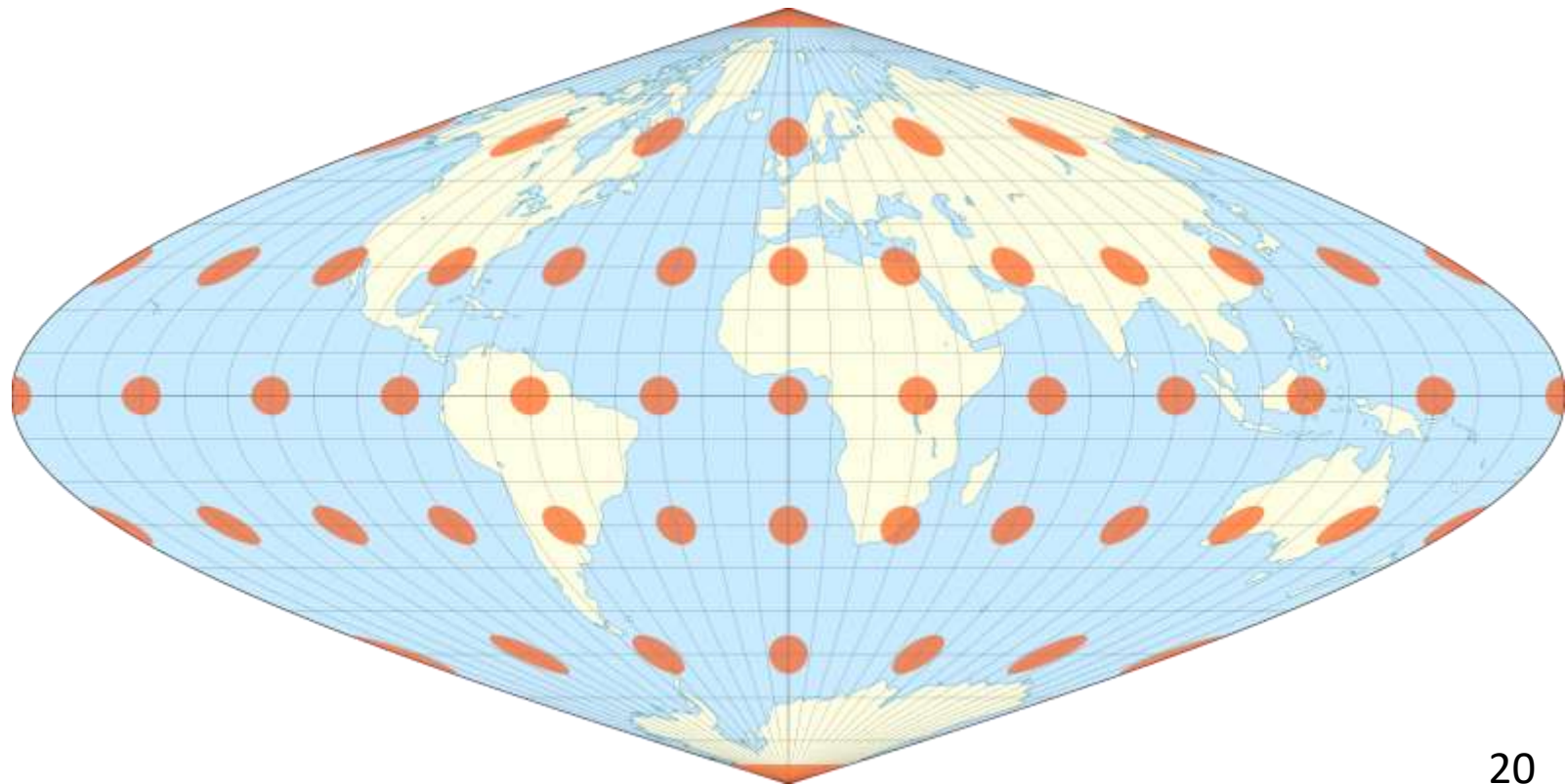
# Sinusoidal Projection

- The **sinusoidal projection** is a simple to compute *pseudo-cylindrical* equal-area projection

$$x := \lambda \cos \phi$$

$$y := \phi$$

- Pseudo-cylindrical:  
Parallels remain straight and parallel
- Prime meridian also remains straight
- Poles represented as points



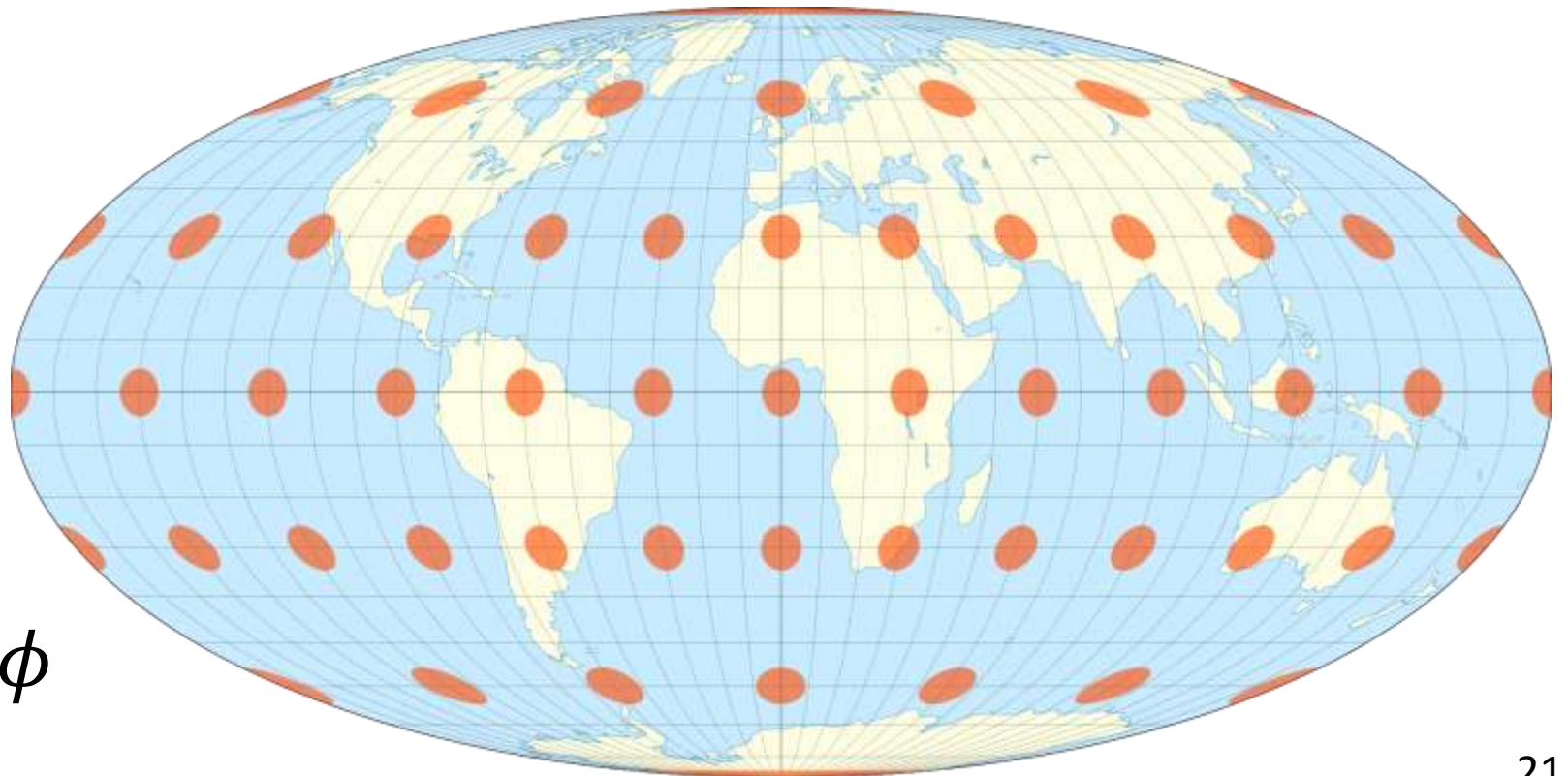
# Mollweide Projection

- The **Mollweide projection** is a pseudocylindrical equal-area projection whose elliptic shape gives a somewhat more natural appearance and reduces distortions

$$x := \frac{2\sqrt{2}\lambda \cos \theta}{\pi}$$
$$y := \sqrt{2} \sin \phi$$

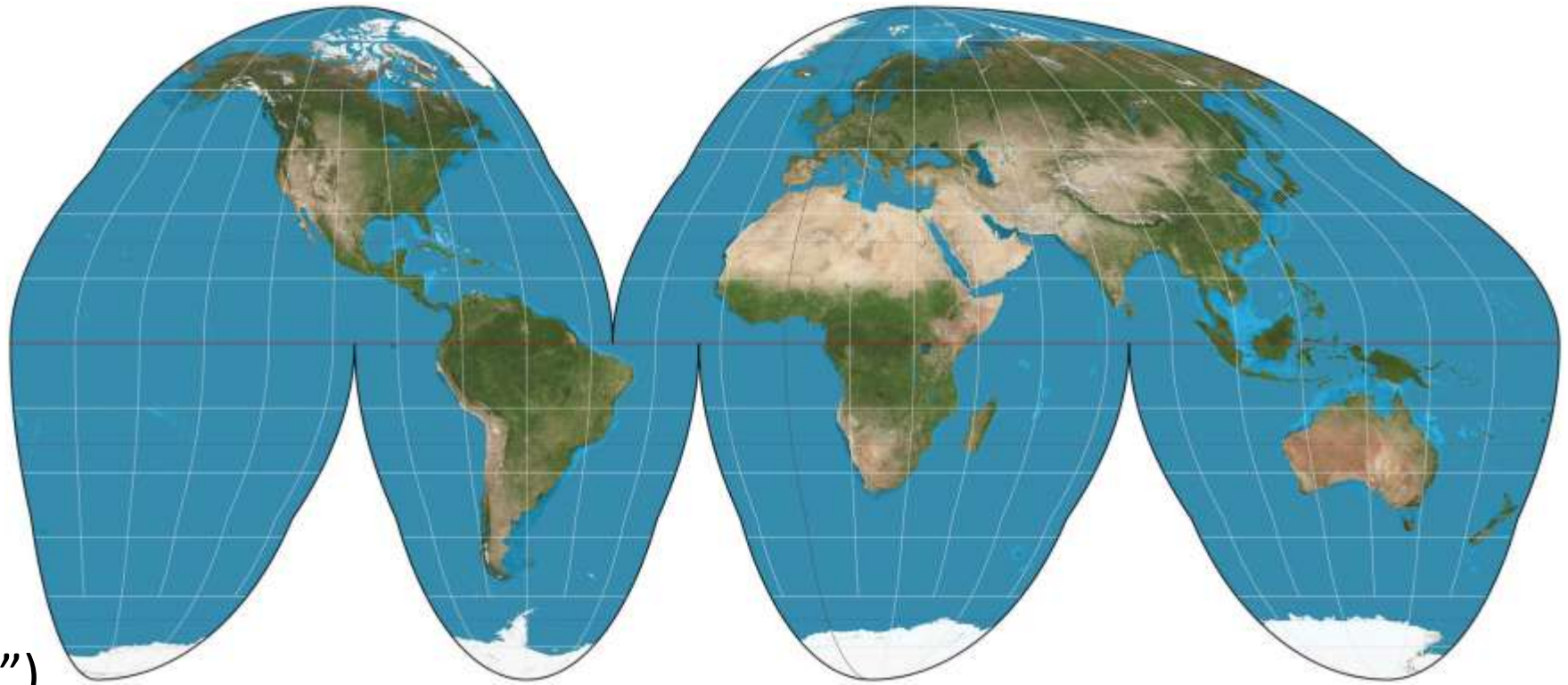
with  $\theta$  such that

$$2\theta + \sin 2\theta = \pi \sin \phi$$



# Interrupted Maps

- **Interrupted maps** split the globe along more than just a single east/west boundary
  - Permits further reduction of distortions in equal-area maps
  - *Example:*  
Goode  
homolosine  
projection
    - equal-area
    - Combines sinusoidal with Mollweide (“homolographic”)



# Summary: Map Making

- **Maps** are produced for various reasons
  - Understanding or illustrating phenomena with a geospatial aspect
  - Navigation
  - cadaster
- All **planar maps** distort angles and/or areas
  - **Conformal** maps (e.g., Mercator) preserve angles
  - **Equal area** maps (e.g., Gall-Peters, Mollweide) preserve areas
- **Pseudo-cylindrical maps** permit equal area mapping with a more natural appearance (most meridians are bent)
  - **Interruptions** can further reduce distortions

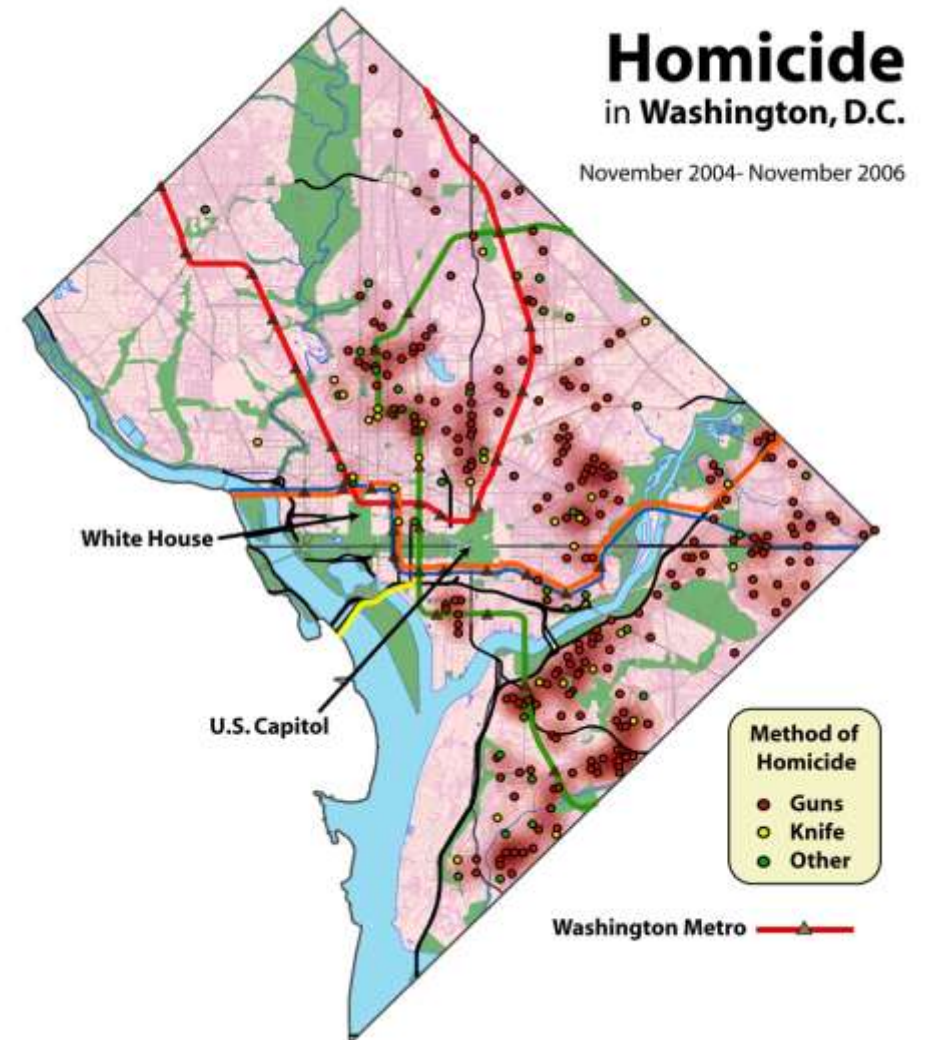


## **Section 6.2:**

# **Point and Line Based Visualization**

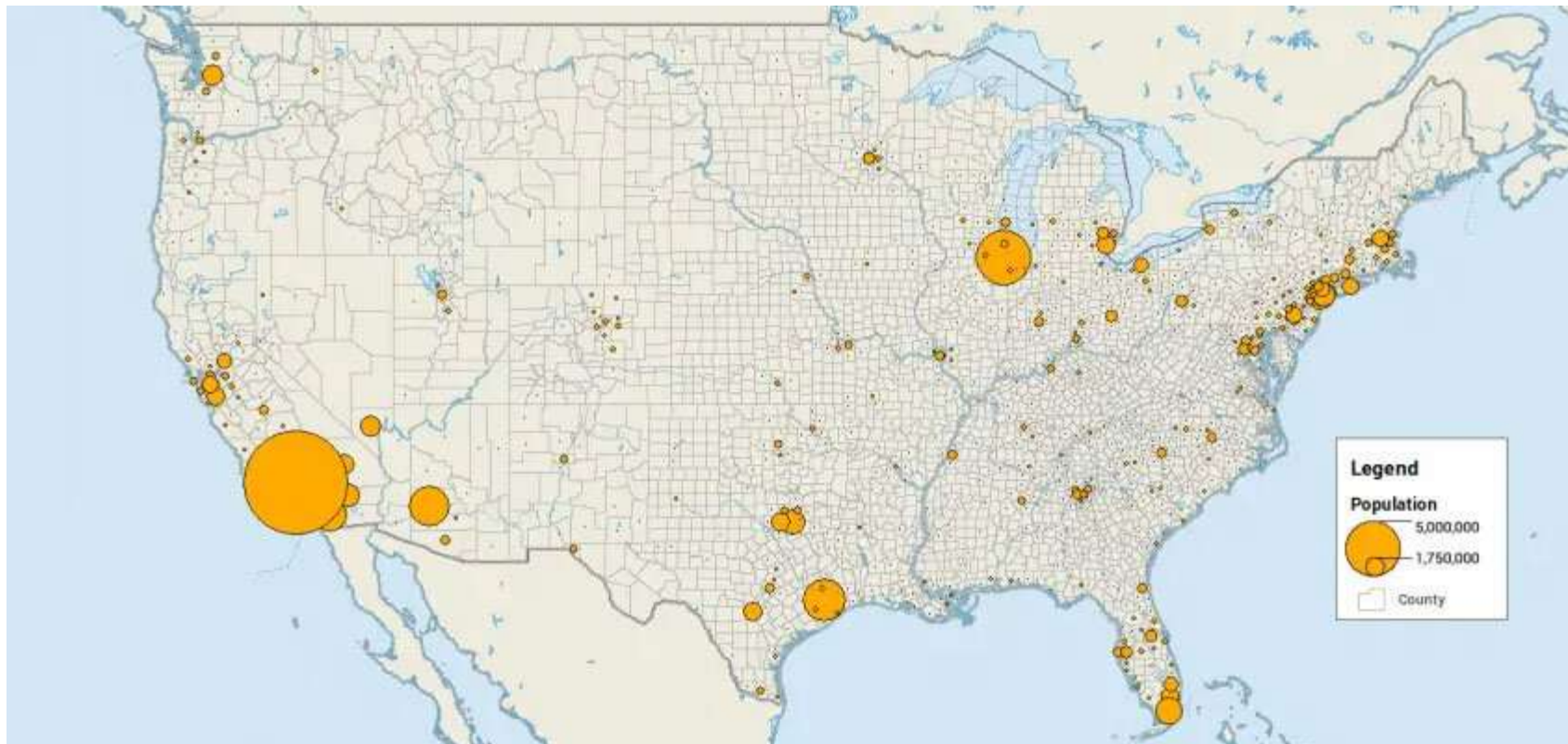
# Point Feature Maps

- **Point data** can arise from
  - intrinsically **discrete** events
    - *Example:* point of sale for a credit card interaction
  - sampling a **continuous** phenomenon (cf. 6.4)
    - *Example:* measurement of temperature at a weather station
- **Point feature maps** directly represent each point with a symbol
  - Most common symbol: Circle
  - Attributes can be mapped to symbol type, size, color



# Proportional Symbol Maps

- In a **proportional symbol map**, symbols are scaled with the absolute magnitude of a quantity at the respective location
  - *Example*: Indicate population of a county via area of a circle
  - *Problems*: Small values are difficult to see, zero values cannot be mapped



# Absolute vs. Apparent Magnitude Scaling

- **Absolute scaling** encodes a quantity  $v$  as the area of a symbol. When using circles,  $A = \pi r^2$ , which results in

$$r = r_0 \left( \frac{v}{v_0} \right)^{0.5}$$

where the radius  $r_0$  of a reference value  $v_0$  decides global scaling

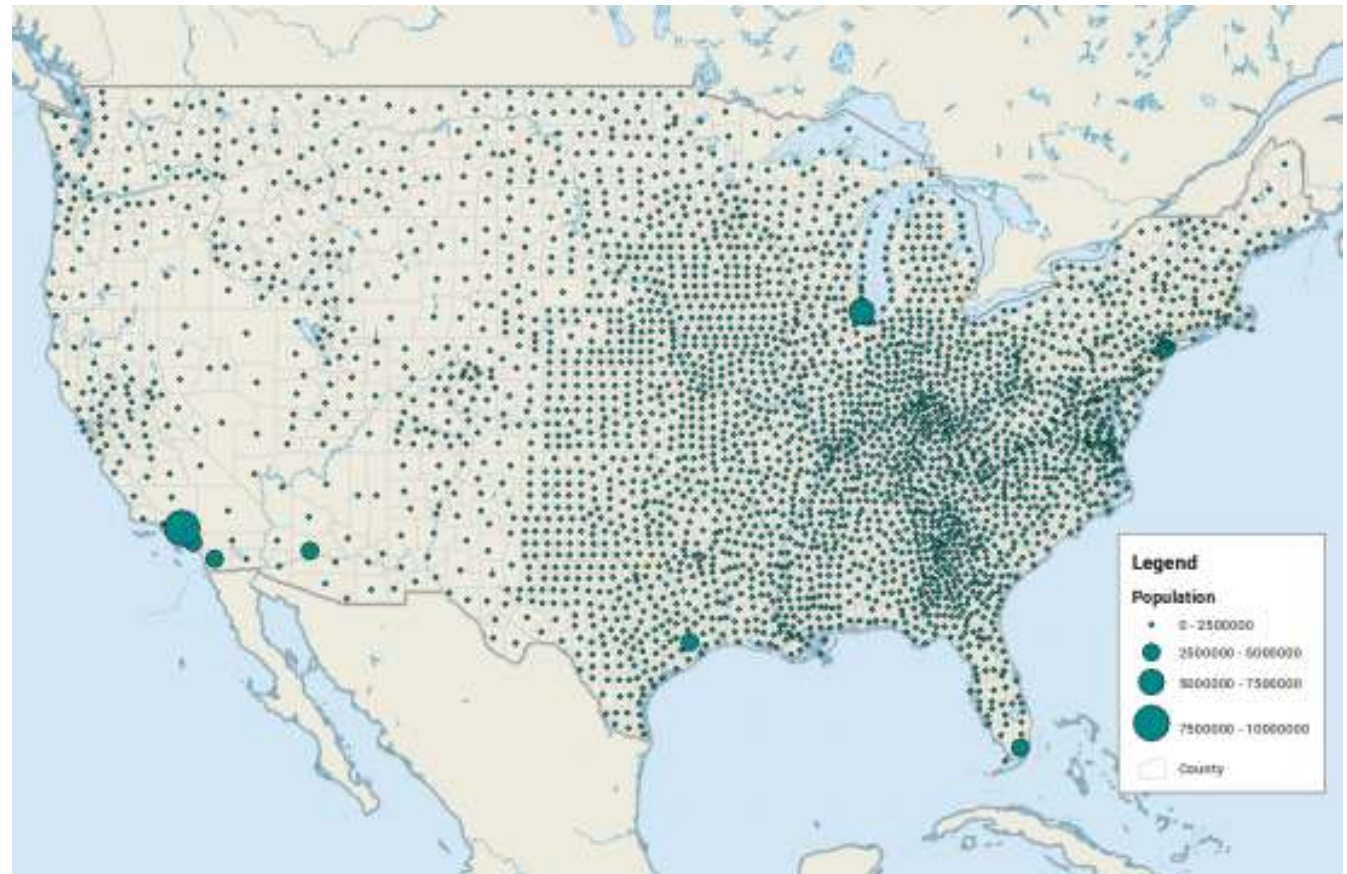
- **Apparent magnitude scaling** (Flannery scaling) compensates for our compressed perception of area (Chapter 5) by slightly increasing the exponent:

$$r = r_0 \left( \frac{v}{v_0} \right)^{0.5716}$$



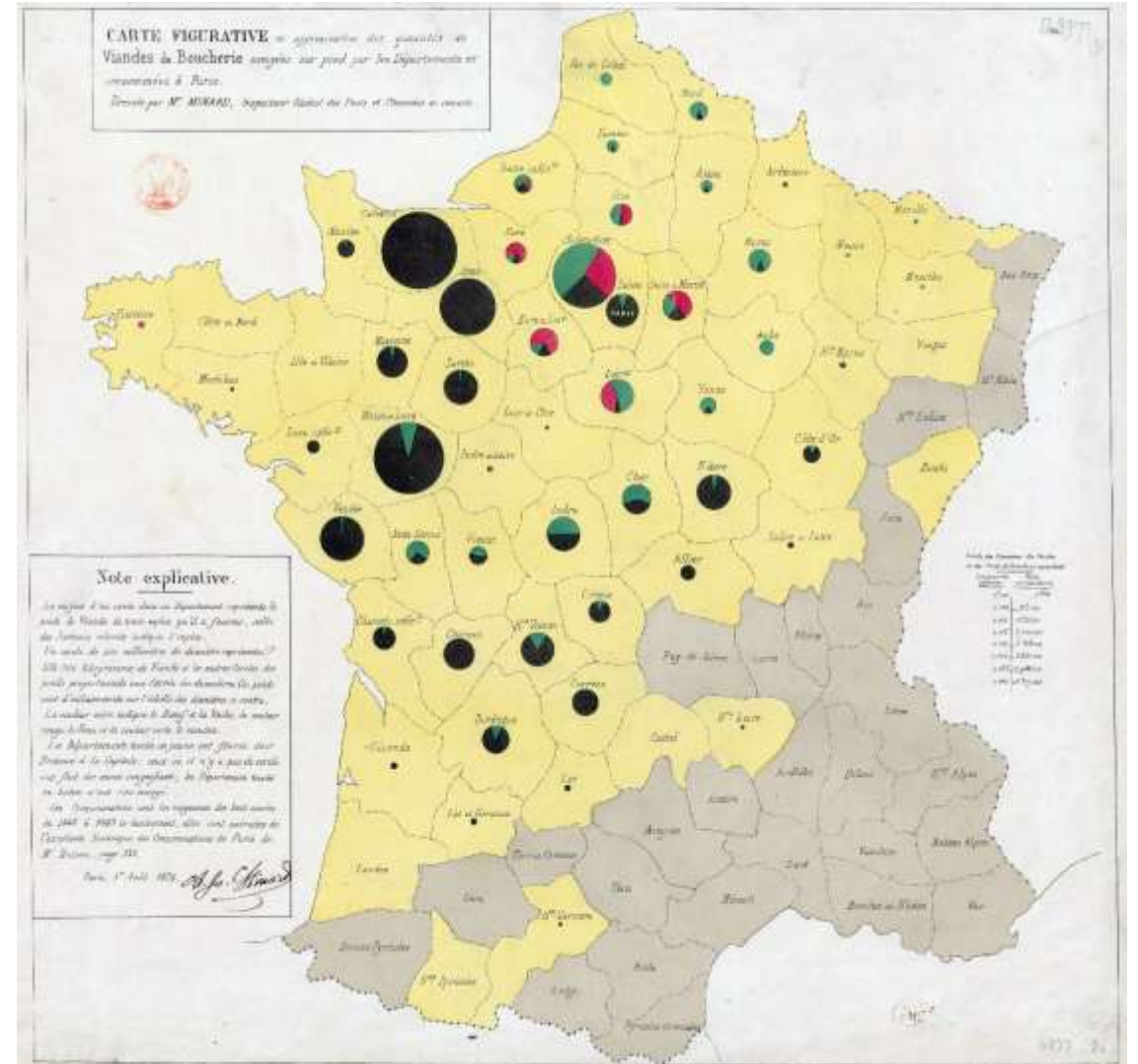
# Graduated Symbol Map

- In a **graduated symbol map**, the quantity is discretized into classes, which are indicated via symbol size
  - *Example:* Indicate population of various US counties
  - In this case, the underlying data is defined at an area level



# Encoding Additional Information

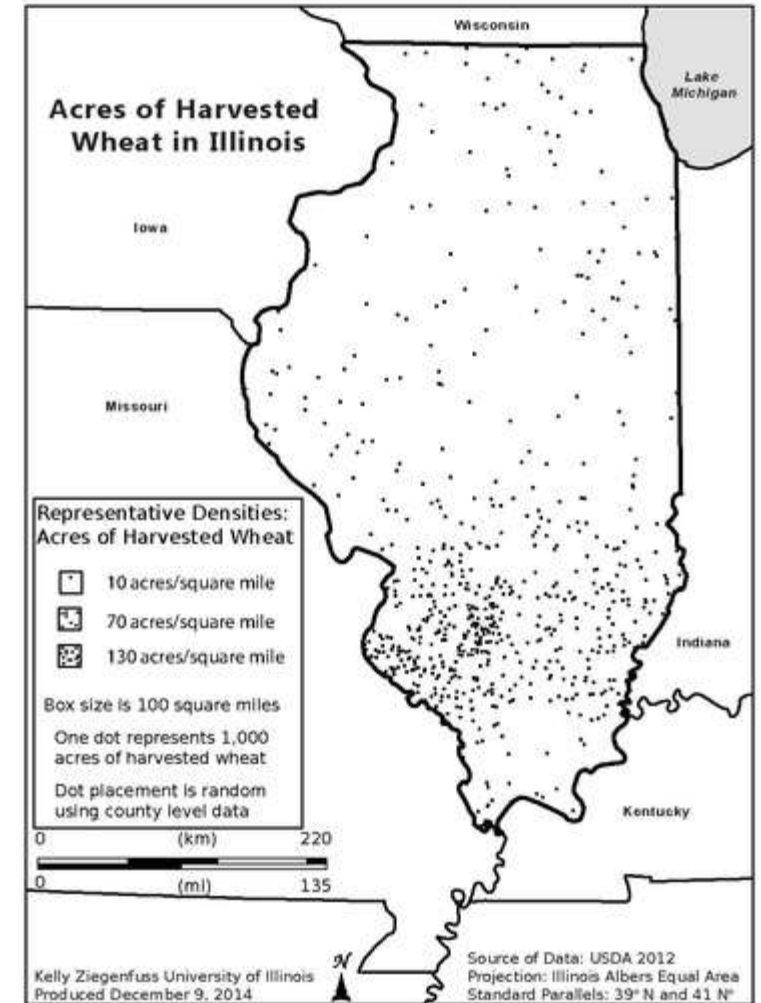
- **More complex symbols** can be used to encode additional information
  - *Example*: Overall amount of meat produced for consumption in Paris, and fraction of type of meat
    - Black = beef
    - Red = veal
    - Green = mutton



Map drawn by Charles Minard (1858)

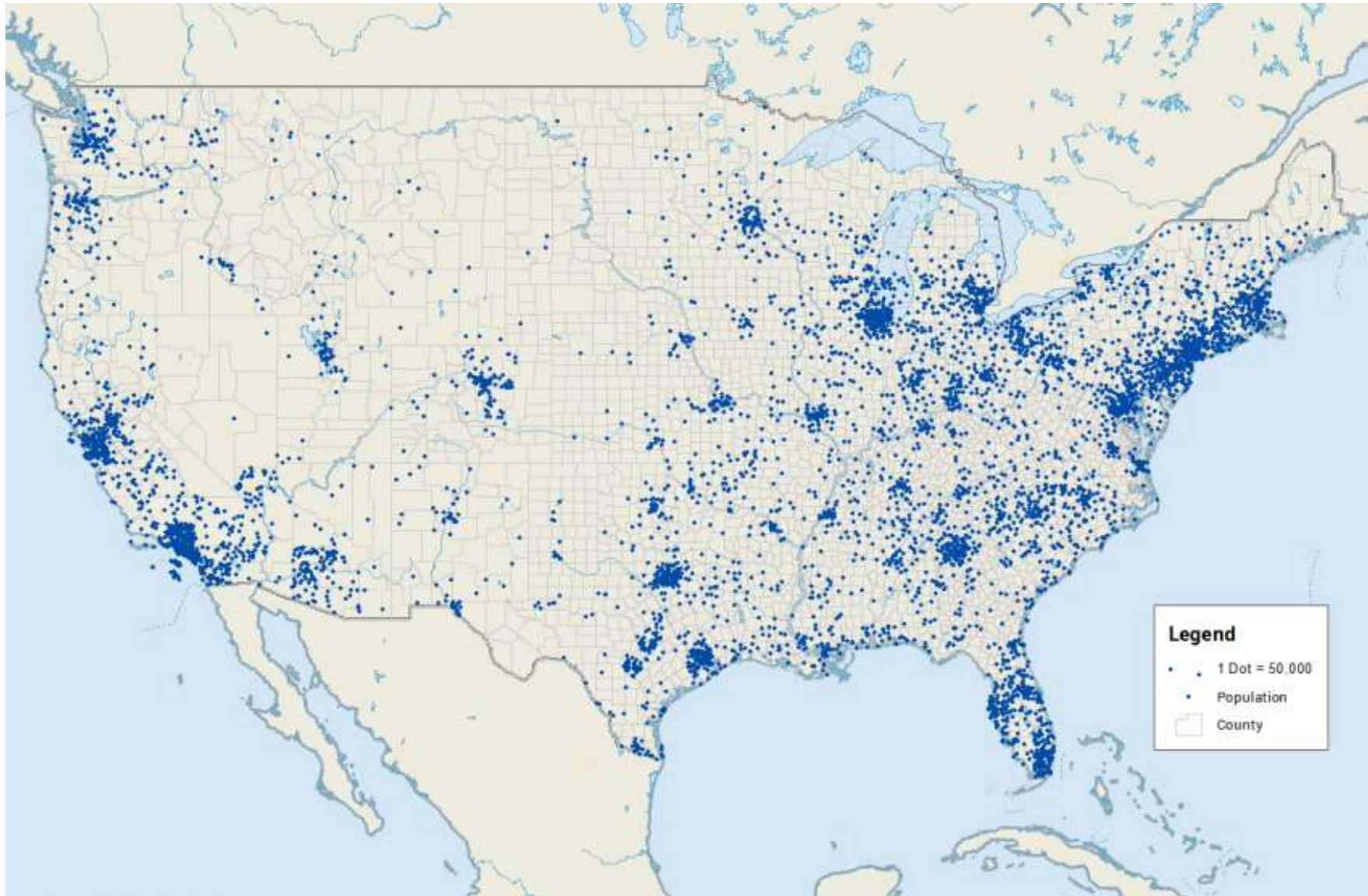
# Dot Density Maps

- In a **dot density map**, each dot represents a certain quantity rather than an individual event
  - Underlying data is typically based on predefined areas, exact placement is more or less random
  - The *dot value* decides how many points will be placed overall and typically correlates with the chosen symbol size
    - These parameters need to be chosen so that excessive overdraw is avoided





## Second Example: Dot Density Map



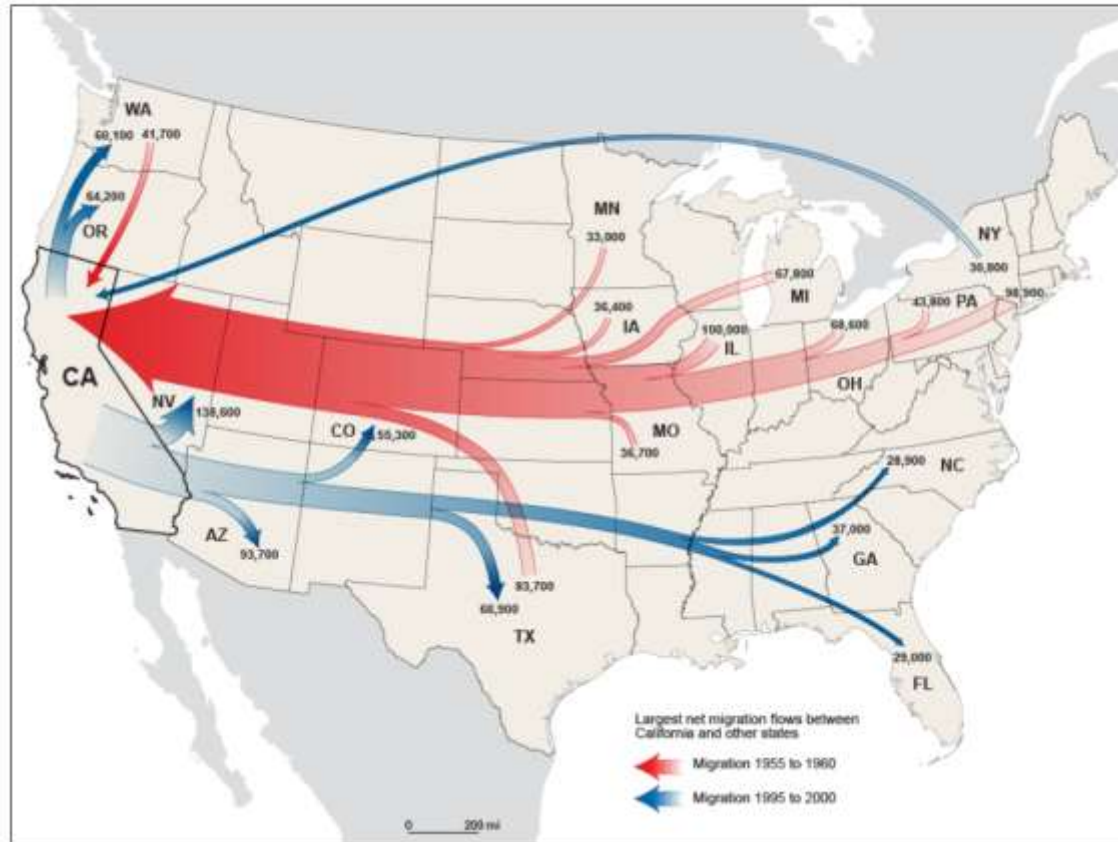


# Flow Maps

- **Flow maps** use curves to visualize movement or transport

Net Migration Between California and Other States: 1955-1960 and 1995-2000

March 7, 2013



(Source: census.gov)



Cargo traffic in Ireland, Harness 1838

# Summary: Point- and Line-based Visualization

- Points in a **point feature map** represent individual items or events at their precise location
- **Proportional symbols** convey a continuous quantity with their size
  - Decide between absolute or apparent magnitude scaling
- **Graduated symbols** encode discrete or discretized values
- Dots in a **dot density map** represent a certain quantity and are distributed over the area for which the quantity was given
- **Flow maps** represent movement or transport as curves

## **Section 6.3:**

# **Area Based Visualization**

# Choropleth Maps

- **Choropleth maps** apply color or shading to pre-defined areas to visualize values related to them
  - Often, areas are political or statistical units
  - “Choropleth” is derived from the Greek words “choros” (place) and “plethein” (to fill)

Total Population of 2000 Census Block Groups

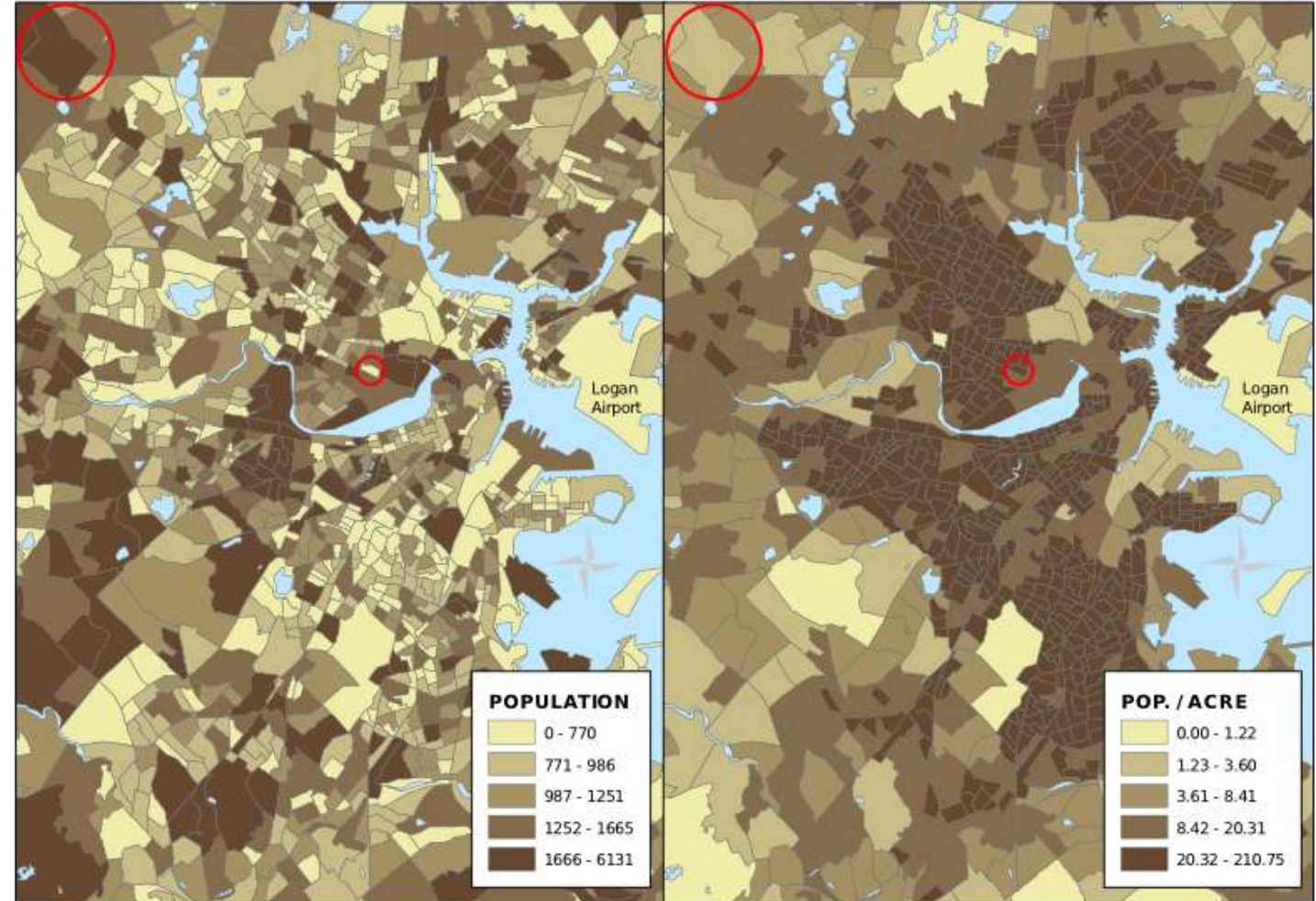




# Choropleth Maps

- **Choropleth maps** apply color or shading to pre-defined areas to visualize values related to them
  - Often, areas are political or statistical units
  - “Choropleth” is derived from the Greek words “chōros” (place) and “plethein” (to fill)
  - Often, quantities are normalized, since areas are visually salient anyway

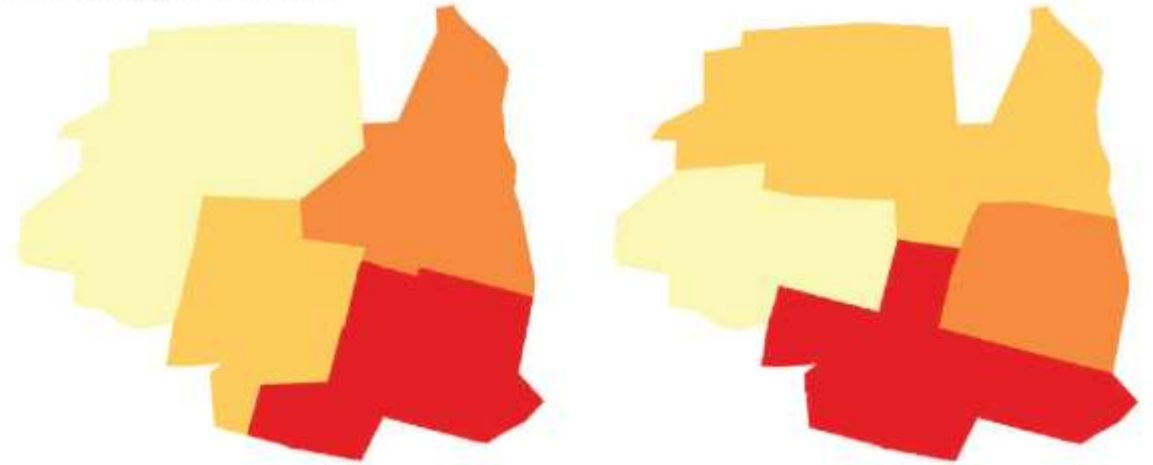
Total Population of 2000 Census Block Groups      Population Density of 2000 Census Block Groups



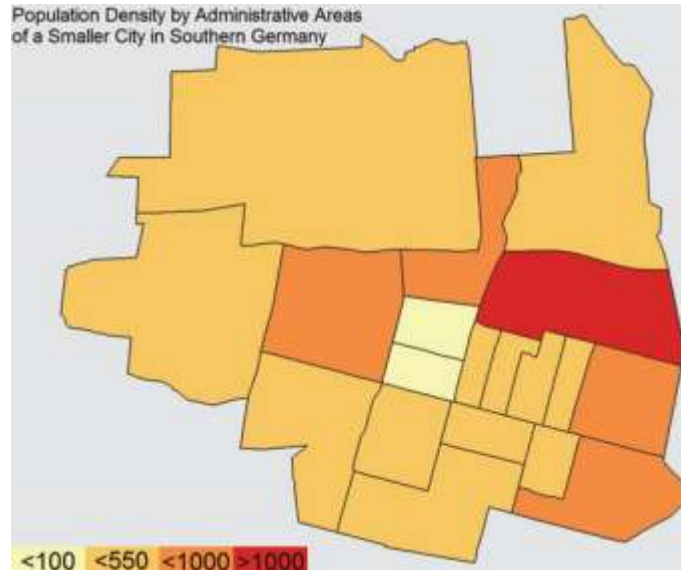
# Effects of Area Aggregation and Classification

- *Note:* The visual impression of choropleth maps can depend significantly on
  - the size and boundaries of underlying areas (top)
  - binning of values (bottom: population density in different districts)

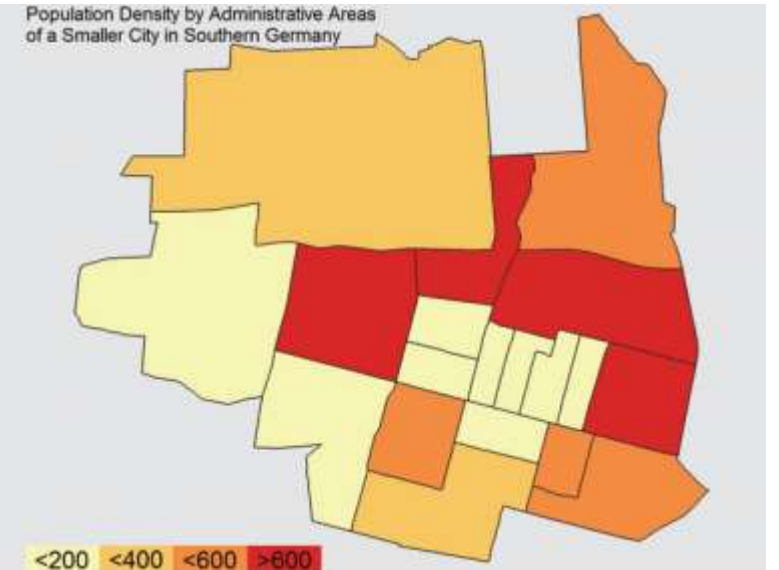
Area Aggregation:



Population Density by Administrative Areas of a Smaller City in Southern Germany

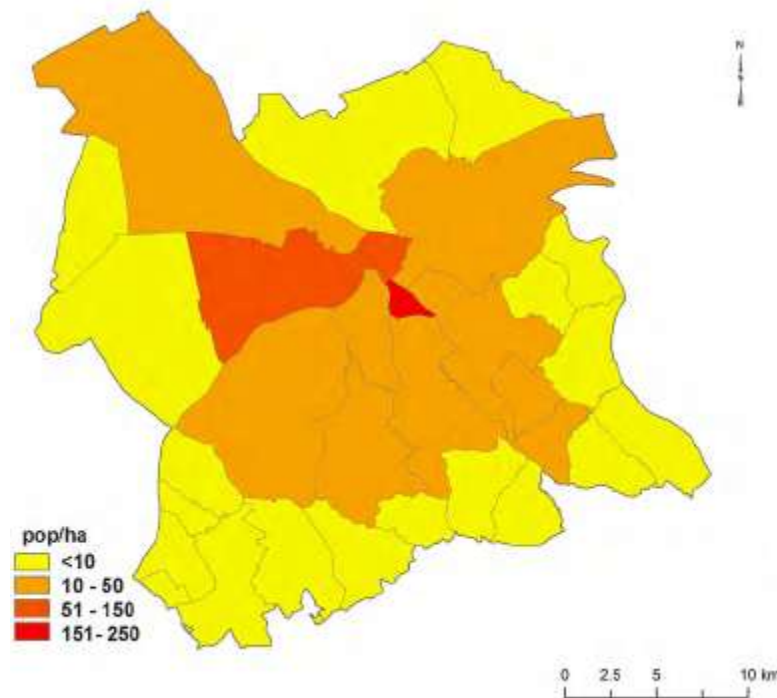


Population Density by Administrative Areas of a Smaller City in Southern Germany



# Dasymetric Maps

- **Dasymetric maps** show densities specifically within the areas in which corresponding values arise

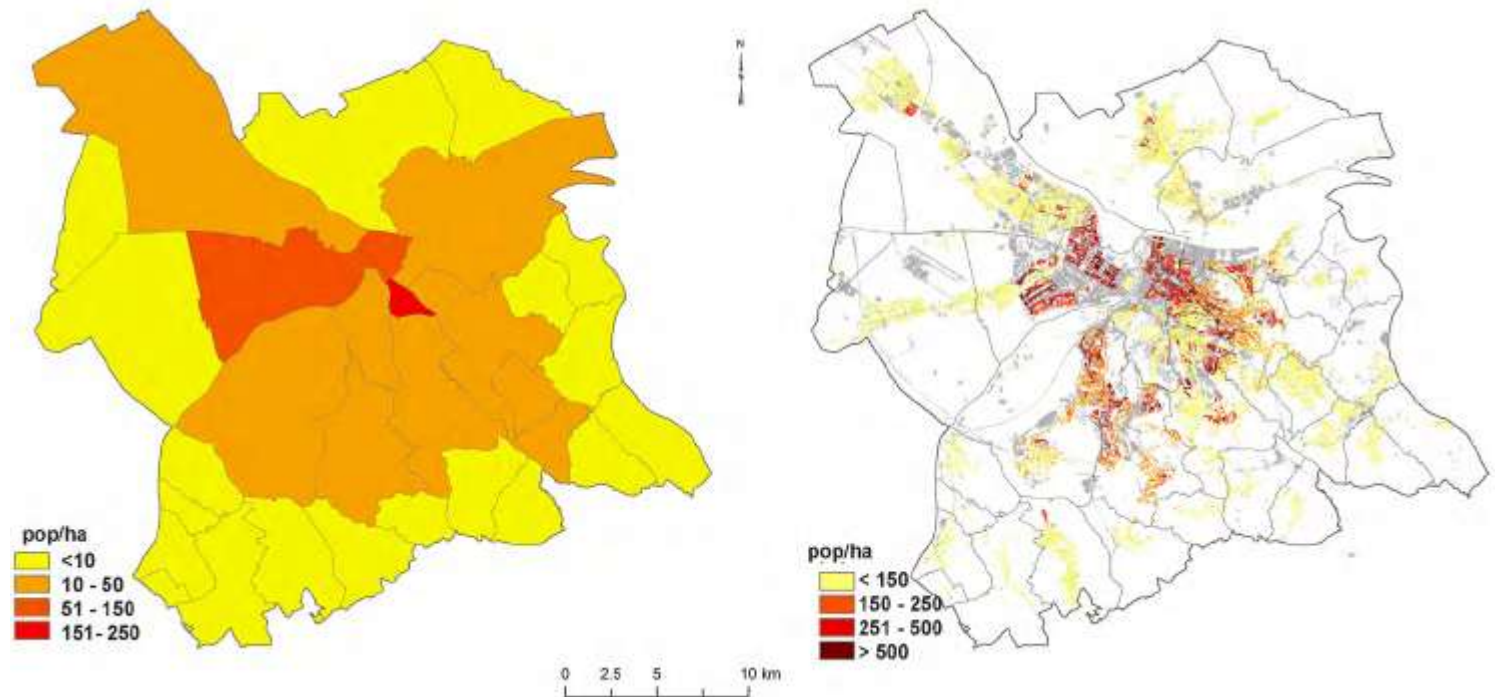


2011 population density in Belgrade [Bajat et al. 2013] 38



# Dasymetric Maps

- **Dasymetric maps** show densities specifically within the areas in which corresponding values arise
  - *Example*: Restricting population density data to residential areas
  - Might re-normalize given values using additional information
  - “Dasymetric” is derived from the Greek words “dasys” (dense) and “metro” (measure)

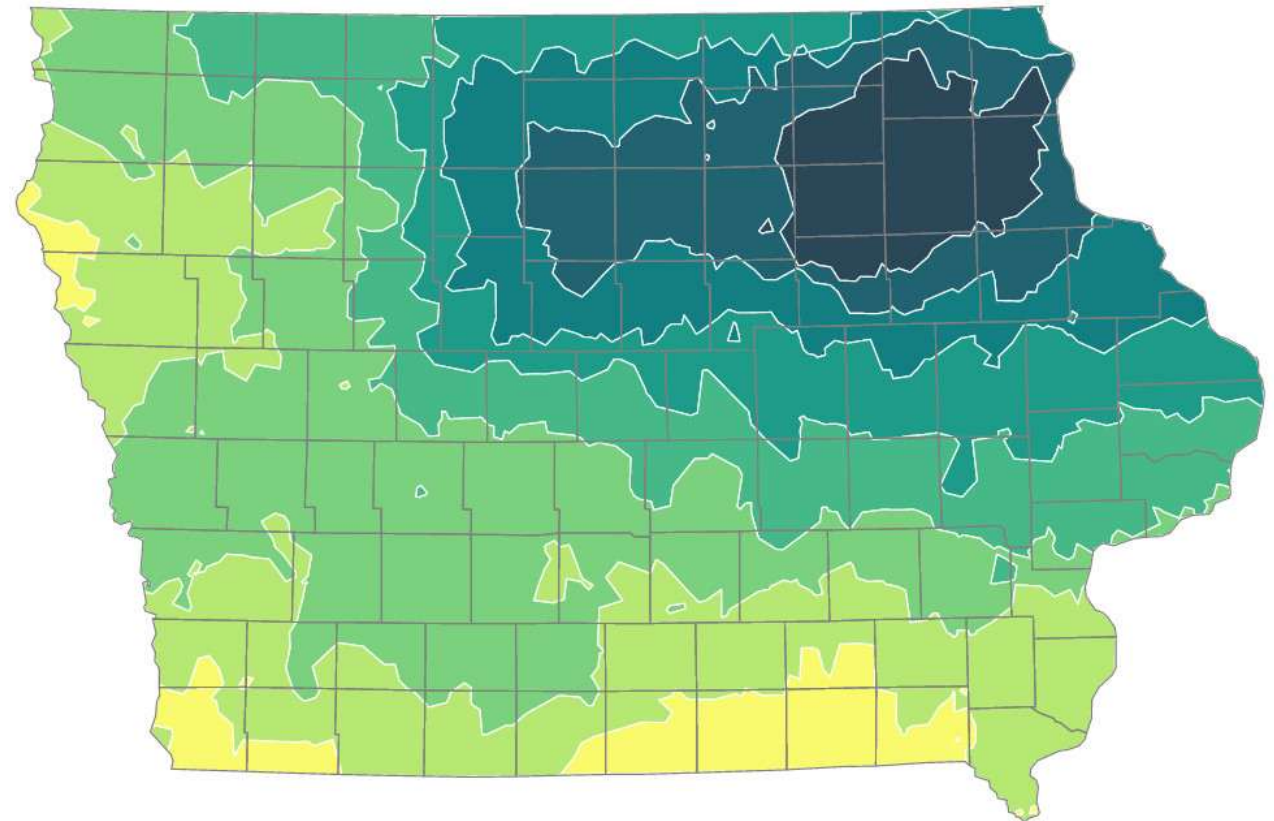




# Isarithmic Maps

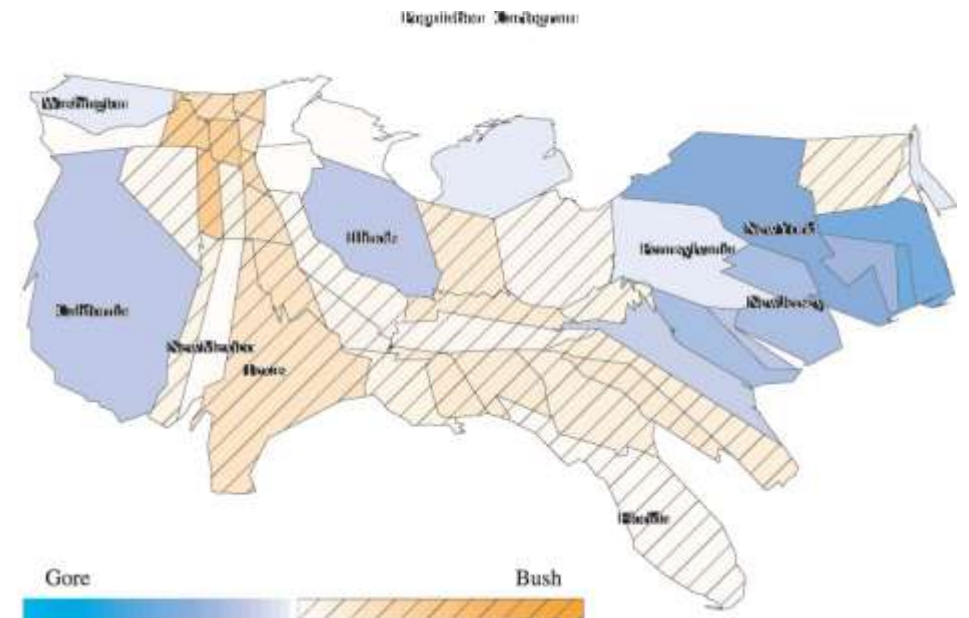
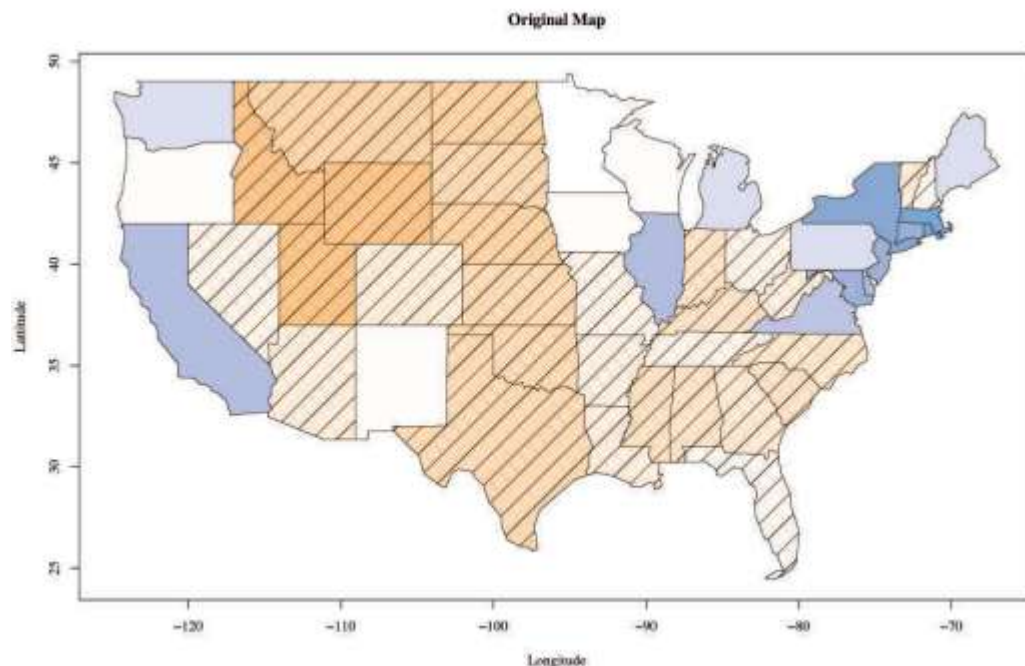
- **Isarithmic maps** include *isarithms* (curves of constant value), often along with color/shading

- *Example*: Amount of precipitation
- Resulting areas reflect the data distribution, are not pre-specified
  - Algorithms in Chapter 9
- Frequently requires scattered data interpolation
  - Section 6.4



# Cartograms

- **Cartograms** distort geographic areas according to a mapped value (most frequently, population)
  - Avoids visual dominance of large areas that might be less relevant
  - Often more natural to map absolute than relative quantities
    - *Example:* Population instead of population density
  - *Example:* Results of the 2000 presidential election



# Noncontinuous Cartograms

- **Noncontinuous Cartograms** scale down individual areas
  - Maintains exact shapes, perfect control over areas
  - Areas become disconnected
    - Rendering them inside the original ones reduces this problem

World Mercator projection with true country size added





# Noncontiguous Cartograms

- **Noncontiguous Cartograms** relax topology (not all adjacency relationships are kept), may also relax shapes



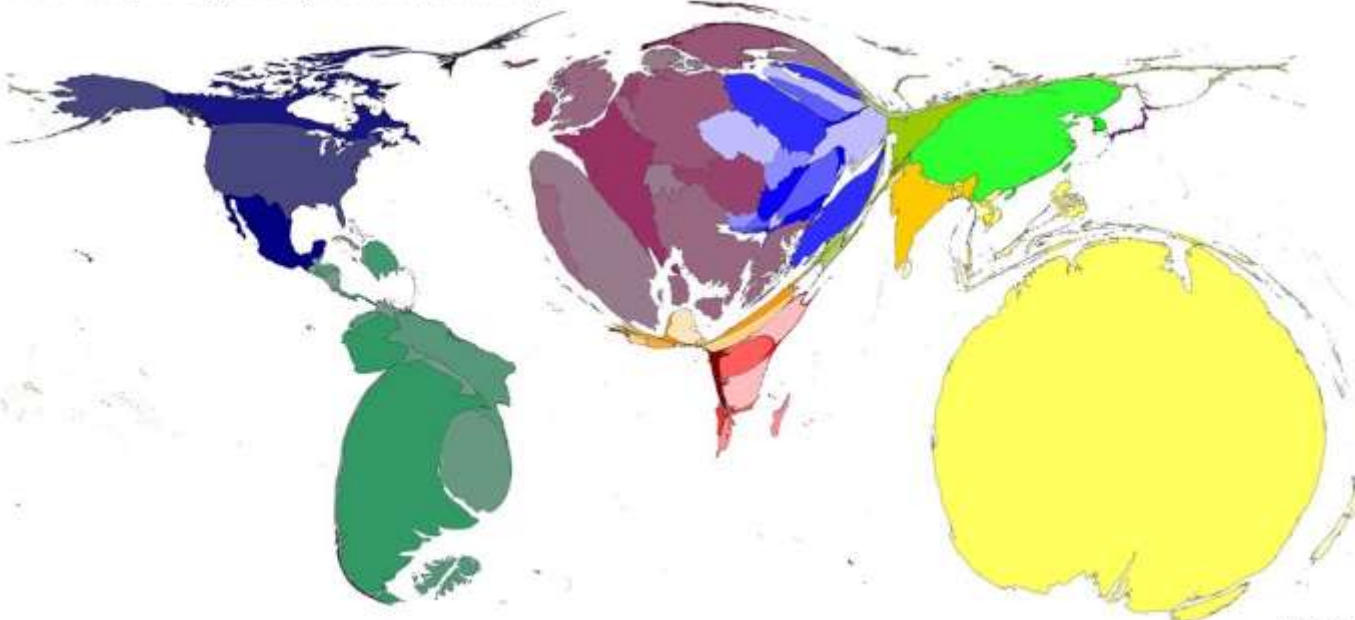




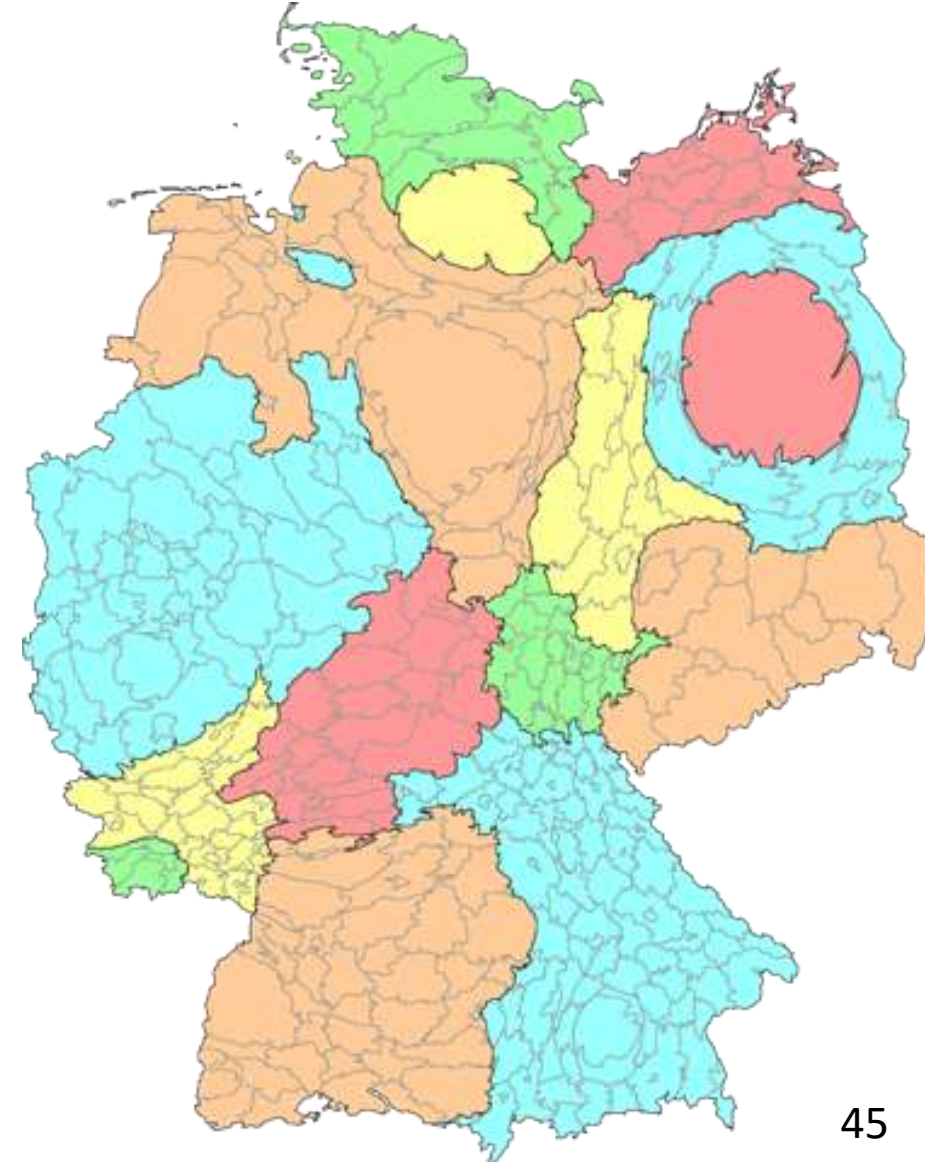
# Continuous Cartograms

- **Continuous cartograms** preserve topology, but distort shapes
  - areas might only approximate the underlying values

World Map of Organic Agriculture (hectares)



Paul & Hennig (2016)



# Summary: Area-based Visualization

- Different types of area-based encodings include:
  - **Choropleth maps** color code predefined areas
  - **Dasymetric maps** refine localization of area data (esp. densities) based on additional information
  - **Isarithmic maps** include level sets of continuous quantities
  - **Cartograms** distort the geometry so that areas reflect some quantity
    - Sometimes even give up topology

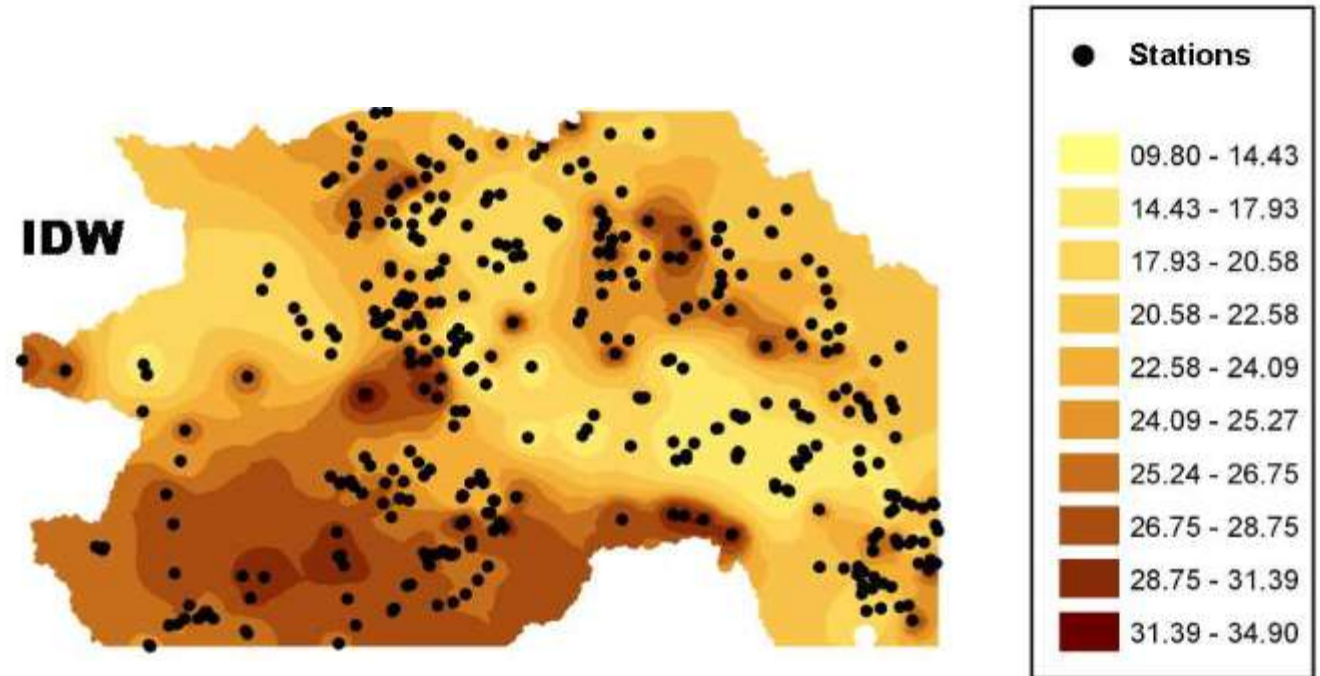
## **Section 6.4:**

# **Scattered Data Interpolation**



# Motivation

- How to reconstruct continuous fields from samples **at arbitrary positions** ( $\mathbf{x}_i, y_i$ )?
  - *Example:* Interpolate between temperatures that were measured at weather stations



Annual average temperatures in Southern Eritrea

# Inverse Distance Weighting (Shepard's Method)

- Shepard (1968) proposes to use weighted average

$$\hat{f}(\mathbf{x}) = \frac{\sum_i w(\mathbf{x} - \mathbf{x}_i) y_i}{\sum_i w(\mathbf{x} - \mathbf{x}_i)}$$

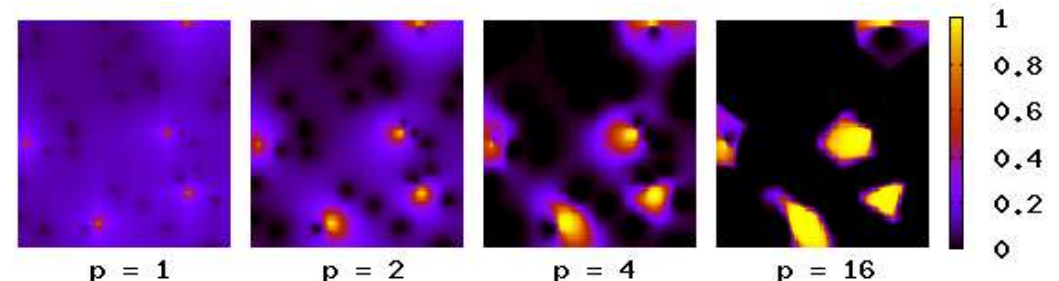
with inverse distance weights

$$w(\mathbf{x} - \mathbf{x}_i) = \|\mathbf{x} - \mathbf{x}_i\|^{-p} \text{ (with positive } p\text{)}$$

- Let  $f(\mathbf{x}_i) := \lim_{\mathbf{x} \rightarrow \mathbf{x}_i} f(\mathbf{x}) = y_i$  to circumvent singularities

- Impact of parameter:  $w(\mathbf{x} - \mathbf{x}_i) = \|\mathbf{x} - \mathbf{x}_i\|^{-p}$

- **Power**  $p$  controls smoothness  
(frequent choice:  $p=2$ )



# Kriging

- **Kriging** (after Danie Krige) also interpolates via weighted averages

$$\hat{f}(\mathbf{x}) = \sum_i \lambda_i y_i$$

but computes weights  $\lambda_i$  by minimizing an objective function

$$\underbrace{E \left[ \left( \sum_i \lambda_i y_i - f(\mathbf{x}) \right)^2 \right]}_{\text{Mean Squared Prediction Error}} - \underbrace{2m \left( \sum_i \lambda_i - 1 \right)}_{\text{Unbiasedness}}$$

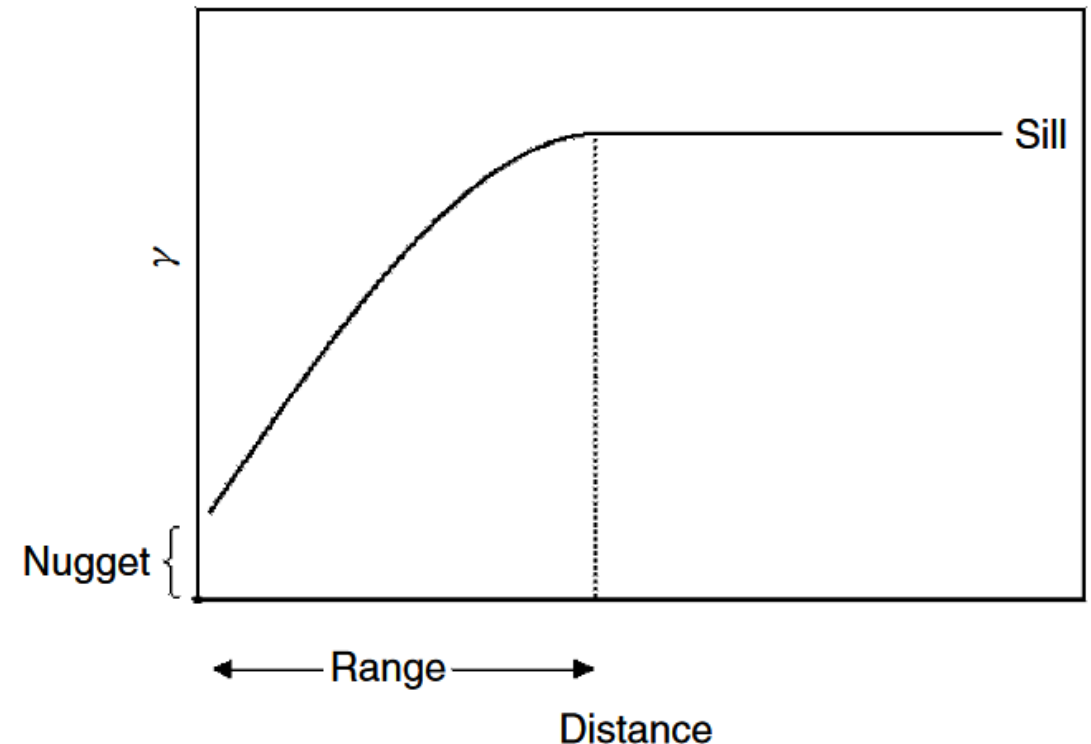
- We consider **Ordinary Kriging**, assumes measurements arise from spatial process  $f(\mathbf{x})$  with fixed but unknown mean  $E[f(\mathbf{x})] = \mu$ 
  - Simple Kriging would assume a fixed and known mean
  - Universal Kriging accounts for polynomial trends in the mean

# Semivariograms

- In order to minimize the expected mean squared error, we need to know the autocorrelation, quantified by a **semivariogram**  $\gamma(\mathbf{x}_i - \mathbf{x}_j)$

$$\text{Var}\left(f(\mathbf{x}_i) - f(\mathbf{x}_j)\right) = 2\gamma(\mathbf{x}_i - \mathbf{x}_j)$$

- In an *isotropic* process, the semivariogram  $\gamma(\mathbf{x}_i - \mathbf{x}_j)$  only depends on distance  $h = \|\mathbf{x}_i - \mathbf{x}_j\|$ , not direction
- A semivariogram is characterized by:
  - The *nugget*, related to the difference between infinitesimally close points
    - Might reflect measurement errors or spatial discontinuity
  - The *sill*, the variance of the process
  - The *range*, a separation distance at which values become uncorrelated





# Semivariogram Selection

- $\gamma$  and its parameters are often selected empirically
  - plot half mean squared differences between values against distances

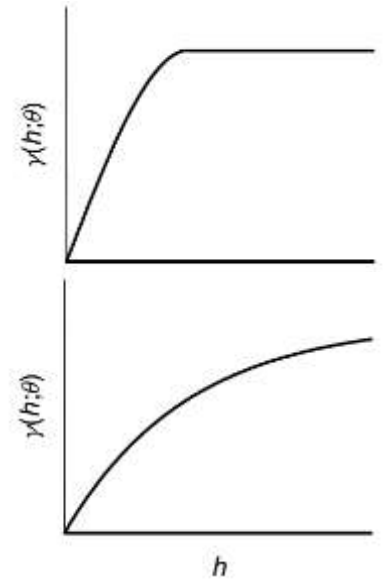
- *Examples:*

- Spherical semivariogram (nugget  $c_0$ , sill  $c_0 + c_s$ , range  $a_s$ )

$$\gamma(h; \theta) = \begin{cases} 0, & h = 0 \\ c_0 + c_s \left[ (3/2)(h/a_s) - (1/2)(h/a_s)^3 \right], & 0 < h \leq a_s \\ c_0 + c_s, & h > a_s, \end{cases}$$

- Exponential semivariogram (“effective range”  $3a_e$ )

$$\gamma(h; \theta) = \begin{cases} 0, & h = 0 \\ c_0 + c_e \left[ 1 - \exp(-h/a_e) \right], & h > 0, \end{cases}$$



# Kriging: Computation of Weights

- Necessary conditions for optimality of objective function

$$\underbrace{E \left[ \left( \sum_i \lambda_i y_i - f(\mathbf{x}) \right)^2 \right]}_{\text{Mean Squared Prediction Error}} - \underbrace{2m \left( \sum_i \lambda_i - 1 \right)}_{\text{Unbiasedness}}$$

yield a linear system:

$$\sum_j \lambda_j \gamma(\mathbf{x}_i - \mathbf{x}_j) + m = \gamma(\mathbf{x} - \mathbf{x}_i) \quad \forall i$$
$$\sum_i \lambda_i = 1$$

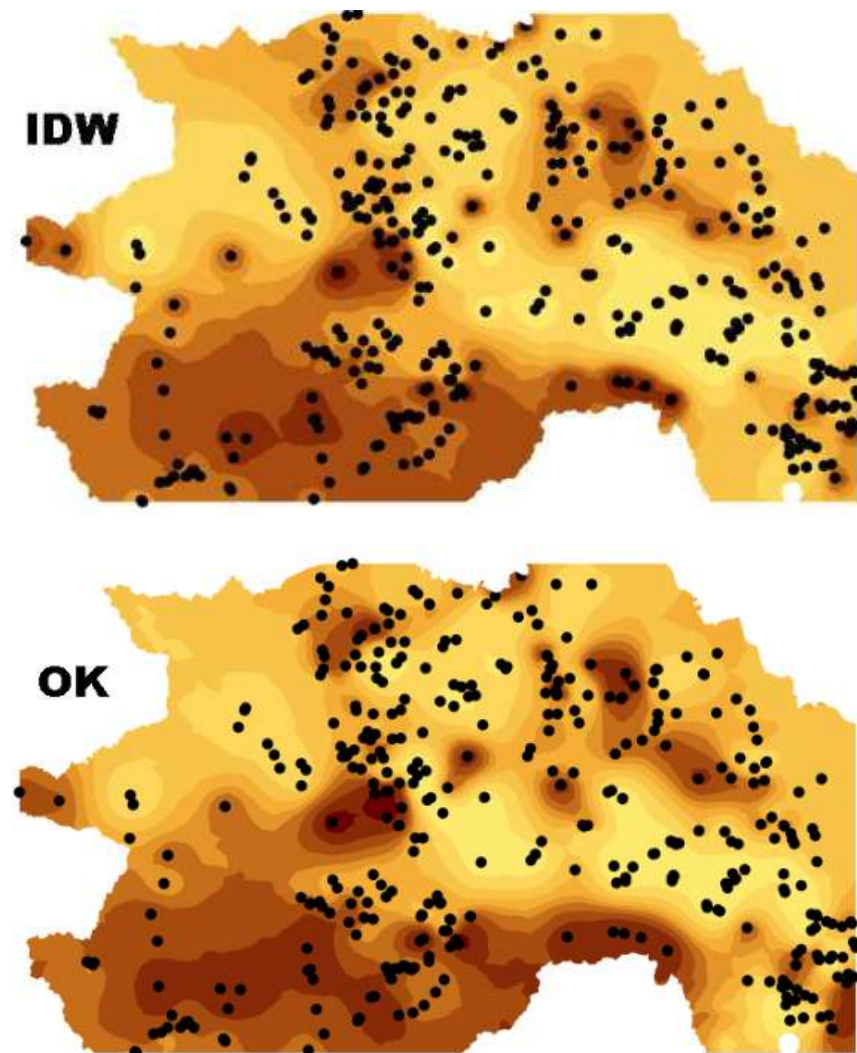
- Note that only the right hand side depends on the target position  $\mathbf{x}$
- Reasonable approximation when only considering the  $k$  nearest neighbors

# Inverse Distance Weighting vs Kriging

- Kriging accounts for distances between measurement sites in addition to distances towards the interpolated point
  - Clusters of nearby measurements carry partly redundant information
- Kriging also yields a variance from which prediction intervals can be computed (assuming Gaussian errors)

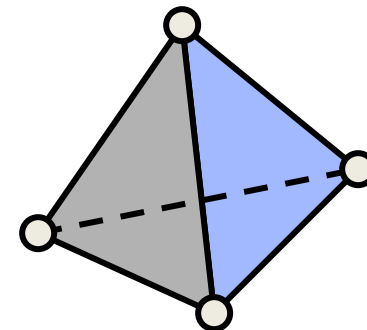
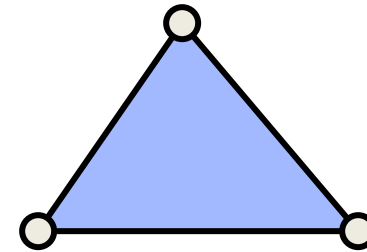
$$\sigma_k^2(\mathbf{x}) = \sum_i \lambda_i \gamma(\mathbf{x} - \mathbf{x}_i) + m$$

- Optimality crucially depends on correct assumptions, in particular, semivariogram



# Linear Interpolation

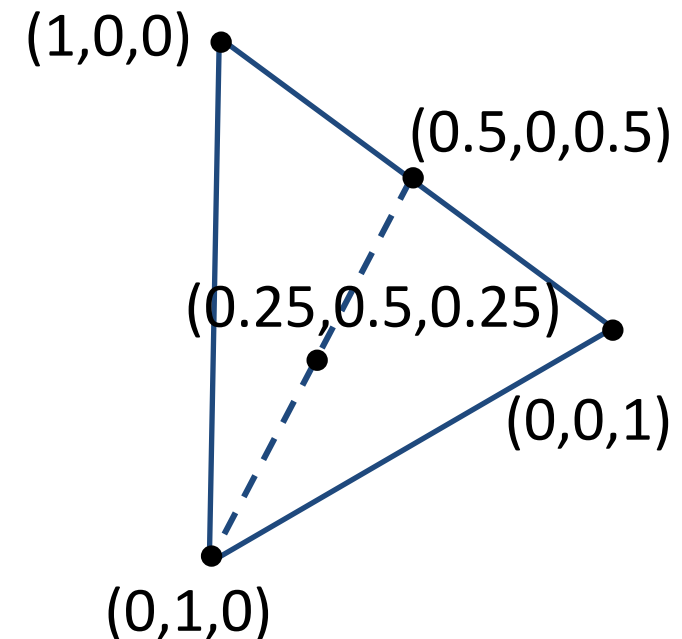
- **Linear interpolation** is fast, easy to implement, and therefore extremely popular in graphics and visualization
- Works on all **simplices**
  - Convex hull of  $d+1$  affinely independent points
  - 1D: *Line*
  - 2D: *Triangle*
  - 3D: *Tetrahedron*





# Barycentric Coordinates

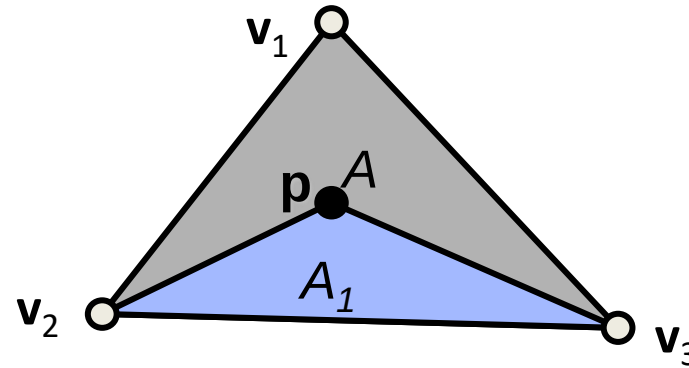
- **Barycentric coordinates** express a point  $\mathbf{p}$  in a simplex relative to its vertices  $\mathbf{v}_i$ 
  - *Intuition:* Place a weight  $\alpha_i$  at each vertex so that the center of mass (“barycenter”) will be at  $\mathbf{p}$
  - To achieve uniqueness, normalize weights so that  $\sum_i \alpha_i = 1$
- Direct way to compute:
  - Solve the linear system consisting of
    - $\mathbf{p} = \sum_i \alpha_i \cdot \mathbf{v}_i$  ( $d$  equations)
    - $\sum_i \alpha_i = 1$



# Geometric Interpretation

- Barycentric coordinate of  $\mathbf{p}$  relative to vertex  $\mathbf{v}_i$  provides the **relative area / volume** of the simplex formed by  $\mathbf{p}$  and  $\mathbf{v}_j$  ( $j \neq i$ )

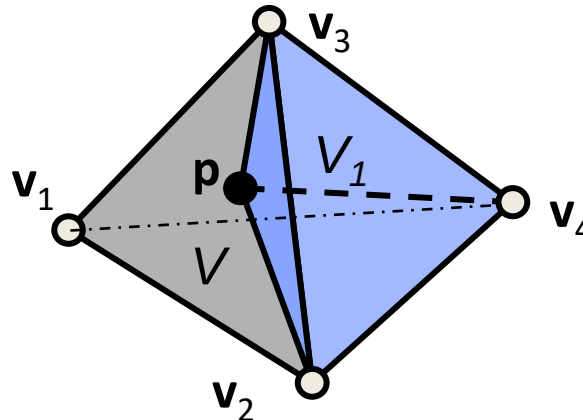
$d = 2$



$$\alpha_1 = \frac{A_1}{A}$$

Opposite  
local area

$d = 3$



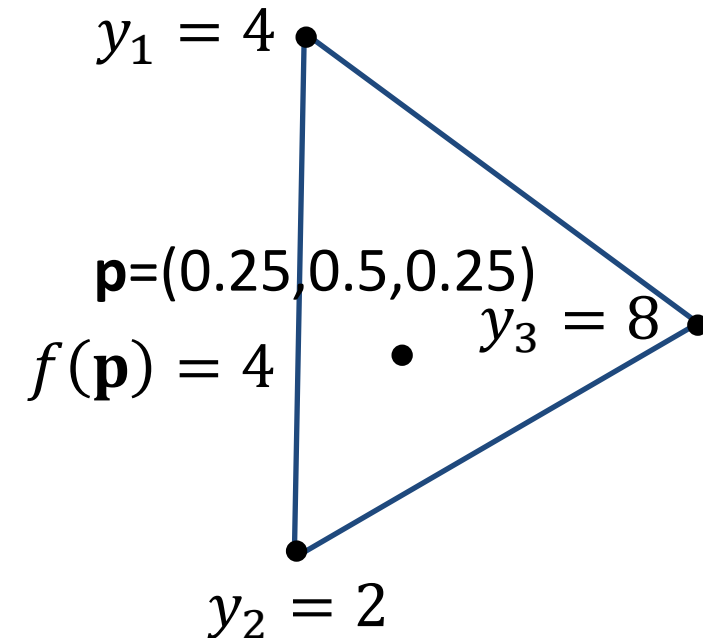
$$\alpha_1 = \frac{V_1}{V}$$

Opposite  
local volume

# Barycentric Interpolation

- **Given:** Simplex with vertices  $\mathbf{v}_i$  and function values  $y_i = f(\mathbf{v}_i)$ ; point  $\mathbf{p}$  inside simplex
- **Barycentric interpolation** computes an interpolated function value  $f(\mathbf{p})$  by
  - computing barycentric coordinates  $\alpha_i$  of  $\mathbf{p}$  with respect to  $\mathbf{v}_i$
  - taking a respective weighted sum of the  $y_i$ :

$$f(\mathbf{p}) = \sum_i \alpha_i y_i$$



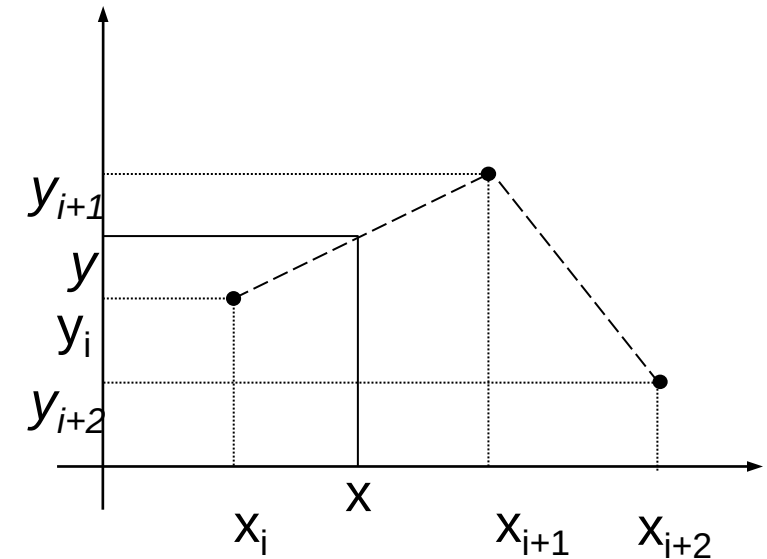
# Piecewise Linear Interpolation

- On simplicial complexes, we can interpolate linearly **piecewise** (over each simplex separately)
  - Simple and fast
  - Disadvantage: Only  $C^0$  continuity between cells
- **Special case:** Univariate ( $d=1$ )
  - Data points:  $(x_0, y_0), \dots, (x_n, y_n)$
  - For any point  $x$  with  $x_i \leq x \leq x_{i+1}$ , evaluate

$$f(x) = (1 - \alpha)y_i + \alpha y_{i+1}$$

where

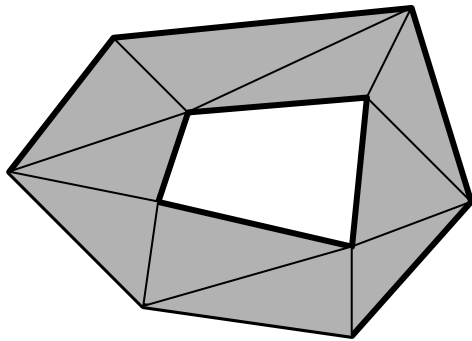
$$\alpha = \frac{x - x_i}{x_{i+1} - x_i} \in [0, 1]$$



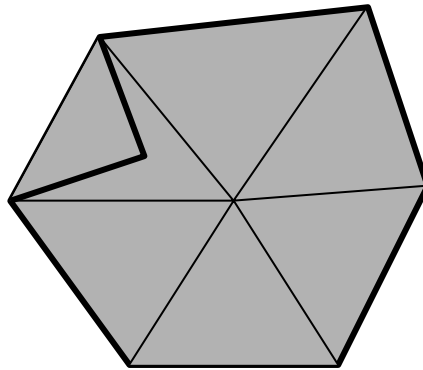


# Triangulating Scattered 2D Data

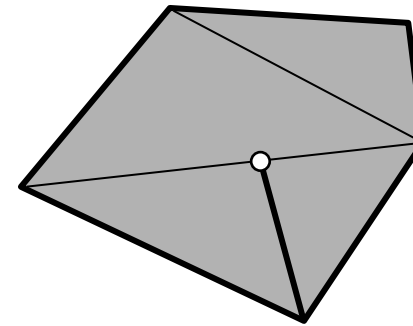
- **Given:** Scattered data points (e.g., in 2D) without any connectivity
- **Goal:** Construct a valid triangulation that can be used for interpolation
  - Examples of invalid triangulations:



Holes



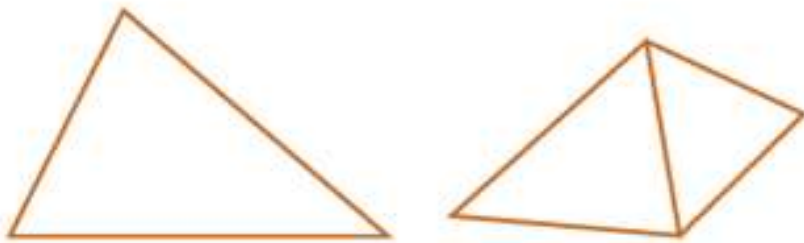
Overlaps



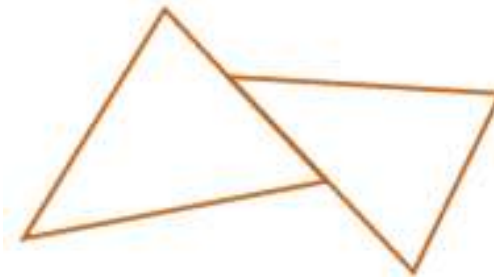
T-Vertices

# Simplicial Complexes

- **Simplicial complex** is a collection  $C$  of points, lines, triangles (etc.) such that:
  - Every face of an element of  $C$  is also in  $C$
  - The intersection of two elements of  $C$  is empty or it is a face of both elements
- Partitions via simplices are called **triangulations**



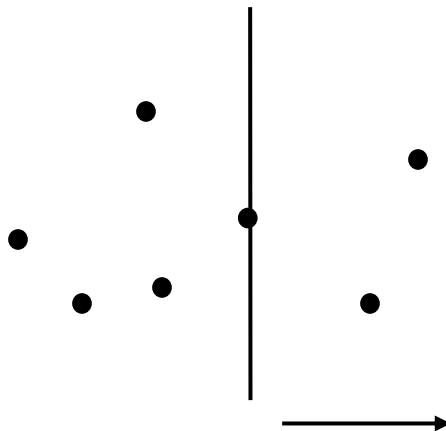
Simplicial complexes



Not a simplicial complex

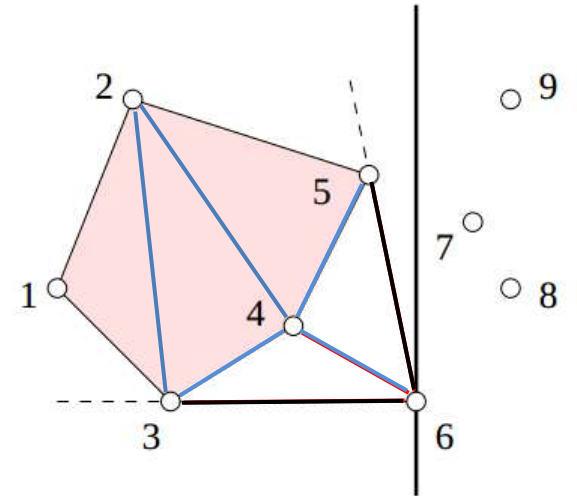
# Plane-Sweep Algorithm

- A simple algorithm for finding a valid triangulation is based on an **imaginary vertical sweepline**, passing from left to right
  - Valid triangulation to the **left of sweepline**
  - Triangles added **at the sweepline**
  - Points **right to sweepline** are ignored



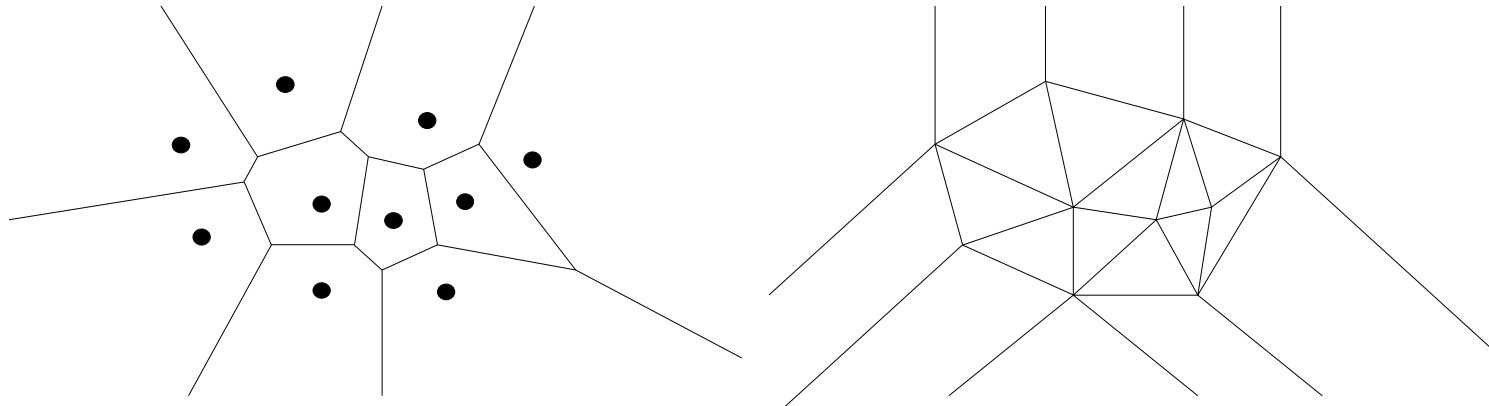
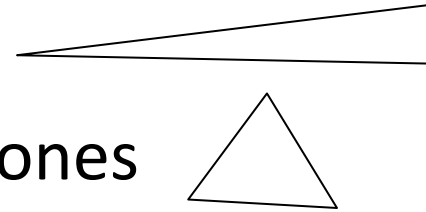
# Plane-Sweep Algorithm: Pseudocode

```
sort points from left to right
construct initial triangle using first three vertices
for  $i=4$  to  $n$  do
    using  $p_{i-1}$  as starting point
        walk counterclockwise along convex
            hull of current triangulation while
            points are "visible", inserting
            edges between  $p_i$  and visited points
    using  $p_{i-1}$  as starting point
        walk clockwise along convex hull of
            current triangulation while points
            are "visible", inserting edges
            between  $p_i$  and visited polygon points
        update convex hull by replacing points visible to  $p_i$ 
            (except tangent points) by  $p_i$ 
endfor
```



# Constructing a “Good” Triangulation

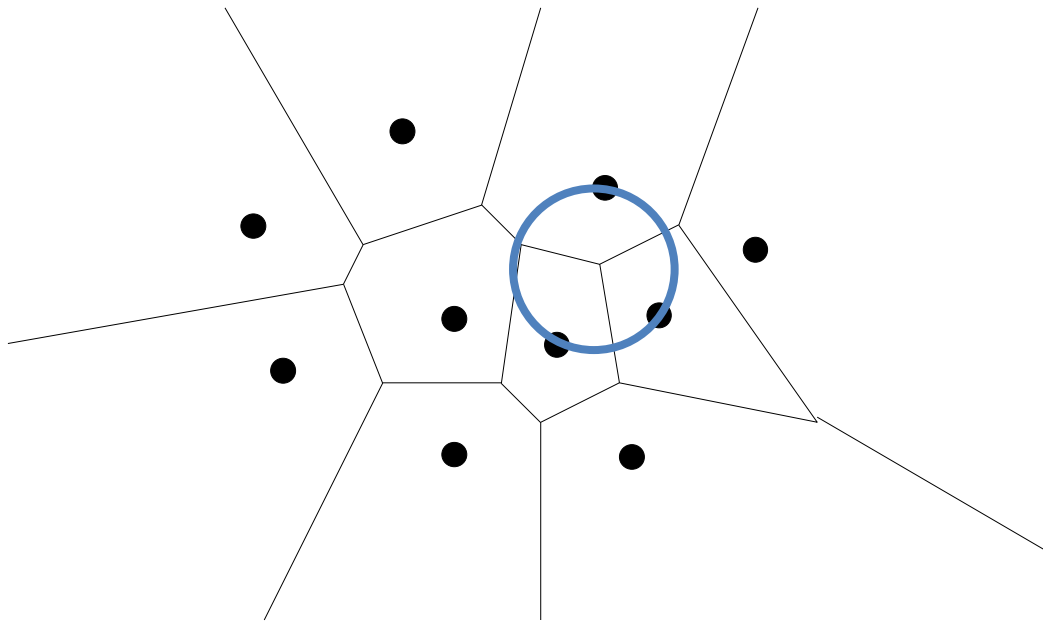
- Plane-sweep creates a valid triangulation, but not always a “good” one
  - A measure for the quality of a triangulation is the **smallest angle** of its triangles
  - Should be maximized to avoid long, thin ones
  - Optimal solution given by **Delaunay Triangulation**, related to Voronoi Diagrams





# Voronoi Diagrams

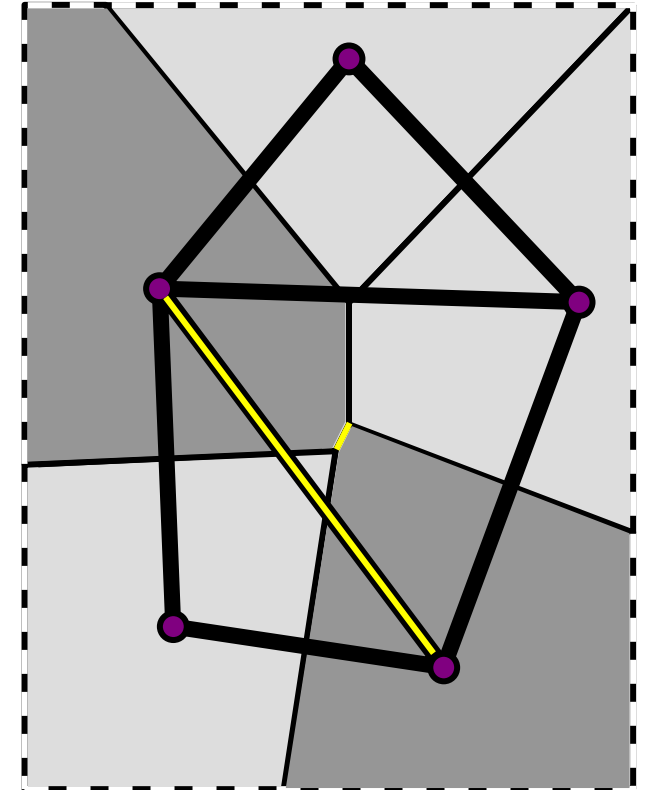
- Voronoi Cells are regions around each sample point (“site”) that covers all points that are closer to that site than to any other site



**Note:** We can center a circle on each vertex of a Voronoi diagram that passes through the three corresponding sites and contains no other site

# Delaunay Triangulation

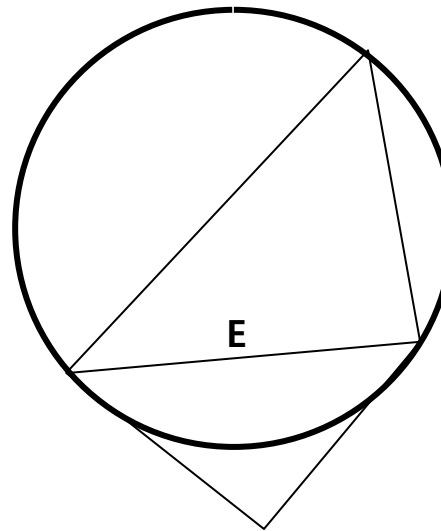
- For points in “general position”, the **Delaunay triangulation** is the geometric dual of the Voronoi diagram
  - Points in  $X$  are nodes
  - Two nodes  $x_i$  and  $x_j$  are connected if the Voronoi cells  $V(x_i)$  and  $V(x_j)$  are adjacent (share an edge)



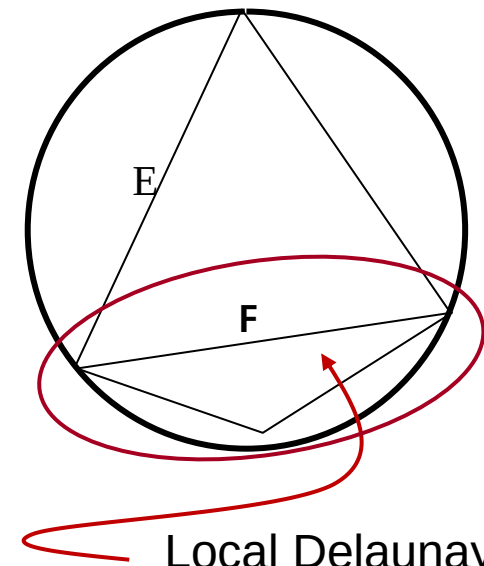
# Local Delaunay Property

Boris Delaunay (1890-1980) characterized edges in a Delaunay triangulation via the **empty circle property**:

- A triangulation is Delaunay iff all edges are locally Delaunay
- An edge  $(x_i, x_j)$  is **locally Delaunay** iff it is on the convex hull or it belongs to two triangles  $(x_i, x_j, x_k)$  and  $(x_i, x_j, x_l)$  so that  $x_l$  does not lie in the circumcircle of  $(x_i, x_j, x_k)$



Local Delaunay met



Local Delaunay violated 67

# Edge Flip Algorithm

Lawson's algorithm to find a Delaunay Triangulation:

**find** an initial (valid) triangulation

**find** all edges where local Delaunay property is violated

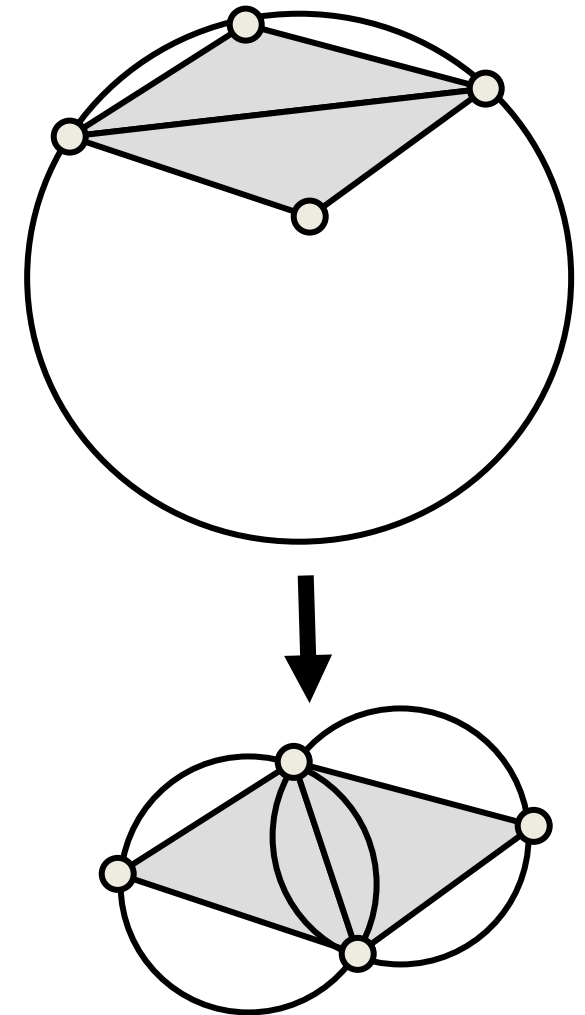
**mark** these edges and **push** them onto a stack

**while** (stack not empty)

**pop** edge from stack and unmark it

**if** (edge does not satisfy Delaunay property)  
        flip this edge

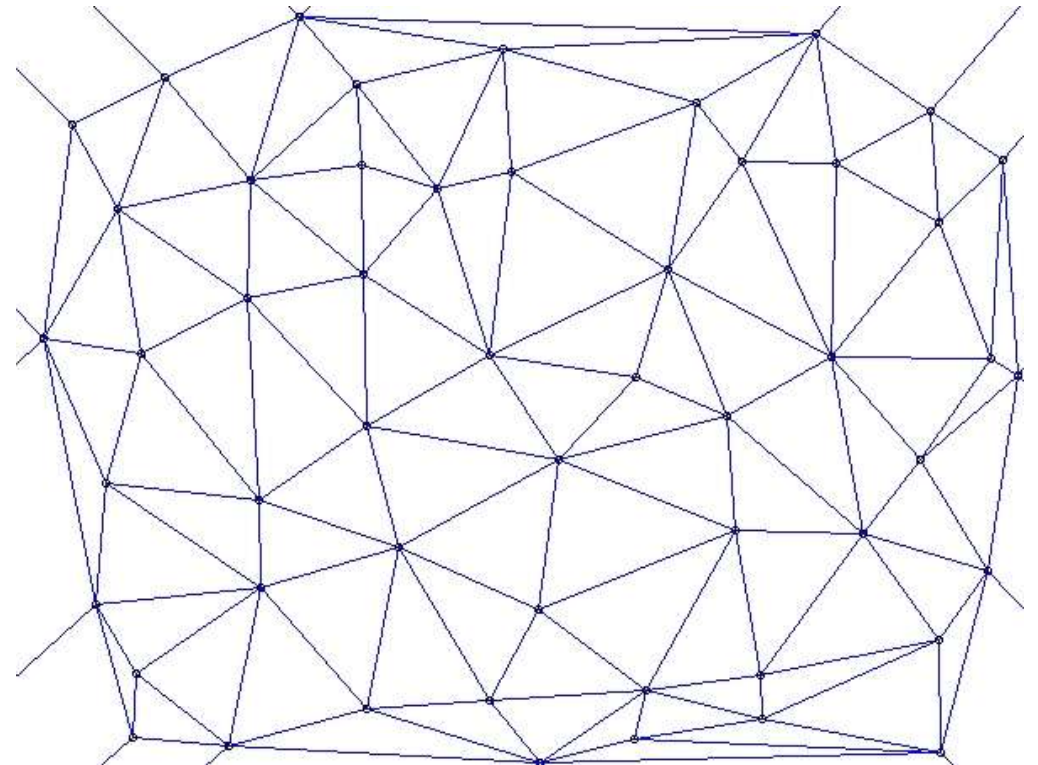
**mark** all unmarked adjacent edges for which the Delaunay property is violated due to the flip and **push** them onto the stack



This algorithm has been shown to terminate in all cases.

# Optimality of Delaunay Triangulation

- Among all valid triangulations, the Delaunay Triangulation maximizes the smallest angle
- Combined with piecewise linear interpolation, it provides an alternative for scattered data interpolation





# Summary: Scattered Data Interpolation

- **Continuous fields** can be reconstructed from scattered measurements via
  - Inverse distance weights (Shepard interpolation)
  - Kriging with suitable semivariogram
  - Linear (barycentric) interpolation after suitable triangulation
- Points in the plane can be triangulated with a **plane-sweep algorithm**
  - Edge flipping can be used to turn the result into a Delaunay triangulation (maximizes the smallest angle)
  - Carries over to Delaunay tetrahedralization in 3D, but algorithms become more complex

# References

- Menno-Jan Kraak and Ferjan Ormeling. **Cartography: Visualization of Geospatial Data**. 4<sup>th</sup> edition. CRC Press, 2021
- Matthew O. Ward, Georges Grinstein, Daniel Keim. **Interactive Data Visualization: Foundations, Techniques, and Applications**. A. K. Peters/CRC Press, 2010
- Lance A. Waller, Carol A. Gutway: **Applied Spatial Statistics for Public Health Data**. Wiley, 2004

## Exam-like Questions

1. State a benefit of the Mercator projection over the Gall-Peters projection. Also state an advantage of the Gall-Peters projection over the Mercator projection.
2. What is Tissot's indicatrix, and what does it tell you about a map projection?
3. What is the purpose of apparent magnitude scaling in the context of proportional symbol maps?
4. Name and describe an approach for scattered data interpolation.